Contracting with Models and Data

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Abstract

Contracting using models and data is a ubiquitous feature of economic interactions, from consulting and information acquisition to portfolio management and delegation to the use of AI-based forecasts to guide decisions. We use a theoretical model to study frictions that arise in these settings. A receiver (she) contracts a sender (he) to help her make a choice. The sender samples data and estimates a "model" mapping choices to outcomes. His model may consistently understate or overstate tradeoffs relative to what the receiver finds plausible, forcing her to entertain multiple models. This creates an incentive for senders who overstate tradeoffs to strategically sample data "locally" in a small neighborhood of a single receiver choice, and for senders who understate tradeoffs to "distantly" sample across the widest possible range of choices. Distant sampling makes the receiver's choices less accurate. This combination of strategic sampling with tradeoff under/overstatement generates gains from *model simplicity*: a sender can make a receiver more optimistic by making her believe his model is not the true model and instead consistently understated or overstated tradeoffs. Allowing the receiver to contract multiple senders only generates full learning if their costs of model provision are low, but requires purchasing multiple models and may be accompanied by an increase in prices.

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1 Introduction

Consider an economist tasked with designing a carbon tax. While she understands the economic tradeoffs underlying taxation, she knows very little about the relevant science and engineering, such as the mapping between emissions and temperature, the long-run trajectory of climate change, or what sorts of manufacturing processes are most pollutive. So, she hires a climate scientist to help her. The scientist collects data, runs complex simulations, and synthesizes his findings into a *model* that illustrates the costs and benefits of different tax rates. The model succinctly maps choices to outcomes that are relevant for the economist.

Yet, learning via a *model* creates frictions. Because the model synthesizes complex data, simulations, and domain expertise, there is no sense in which the economist can substantively audit how the scientist fit the model, such as qualitative features of the data he used. More broadly, the scientist and economist have different ways of approaching the problem. The scientist's model may take an overly *simplistic* view relative to the economist. He may mechanically overweight ecological spillovers the economist views as second-order, or neglect political backlash from high tax rates. The economist could seek additional opinions, but this is costly, and she may not necessarily learn more. Thus, while contracting is valuable, uncertainty about *how models are formed* shapes the quality of information.

This illustrative example reflects a broader set of economic interactions where a principal contracts an agent to provide expert analysis that guides her decisionmaking. A CEO may enlist a management consultant to guide workforce restructuring, an investor may hire a portfolio manager to inform investment decisions, or a voter may ask a politician to justify a policy with evidence from a pilot. In all these cases, the agent does not present the principal with raw data but with a *model* —a synthesis of an agent's expertise, knowledge, and the data he has access to. These models describe how the receiver's actions — layoff decisions, stock choices, or tax rates — map onto outcomes — profits, financial returns, or voter wellbeing. Consultants, for instance, routinely present clients with a series of "if-then" scenarios that describe how a firm's profits vary with the intensity of layoffs, or how investment returns depend on portfolio choices.

These sorts of settings have become particularly relevant with the proliferation of *AI-agent based modelling*: an AI program may help any of the principals above study patterns in data. However, these models are ultimately blackboxes: they often rely on highly nonparametric methods to fit data upstream, and the principal often has no idea what kind of data a model was fine-tuned or trained on.

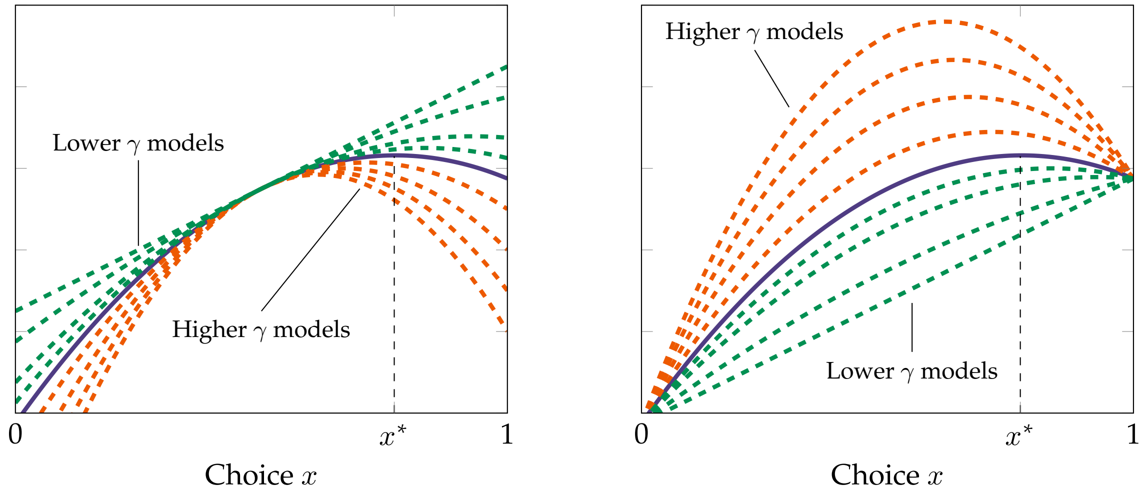
This paper theoretically studies frictions that emerge when contracting with models. A receiver (principal) needs to make a choice. She contracts a sender (agent) to provide information, who would like to make the receiver optimistic. The sender gathers data to estimate a *model*: a data-generating process (DGP) mapping actions to outcomes. However, the model may be *simplified* in that it neglects key tradeoffs, or emphasizes risks the receiver thinks are unimportant. Moreover, the receiver does not observe the data the model was estimated on.

I address four broad questions. First, if the sender estimates a simplified model, how does the receiver learn about the true DGP, and how does this depend on beliefs about the *data* it was estimated on? Second, what kinds of data does the sender prefer to gather? Third, can a sender benefit from supplying models that deliberately understate or overstate tradeoffs? Finally, how are these distortions shaped by *competition* among a market of senders?

Model Overview A receiver — a CEO, investor, or voter — must choose an action x to maximize an outcome y — such as profits or the value of a public good. She believes that the relationship between x and y is given by a DGP $y = \alpha_0 + \alpha_1 x - \gamma x^2$ for $\gamma \geq 0$. This captures a tradeoff between a linear benefit and a convex cost, with γ representing the intensity of this tradeoff. She thus hires a sender to acquire information for her. The sender samples x values from a distribution and observes the corresponding y values, producing data (y, x). He may "locally" sample information about x values in a small neighborhood of a point, or "distantly" sample information about x values far across the choice space. Formally, the sampling choice is characterized by both a "mean" and a "variance" term. His sampling choice is unobserved by the receiver.

The sender then fits a model to the data. He considers similar models to the receiver, except that the set of γ values available to him is a subset of the receiver's, limiting his ability to model tradeoffs. Sender types are heterogeneous and differ in the range of γ values they can fit models to. We say a sender's models *understate* tradeoffs if the set of γ values they consider tend to be *lower* than the ones the receiver considers. They *overstate* tradeoffs if the γ values they consider tend to be *larger*. The receiver observes the sender's model, updates her beliefs over the true DGP, and makes her decision. The sender seeks to maximize the receiver's posterior expectation of y. For example, a portfolio manager may receive a cut of an investor's profits, a consultant's bonus may depend on his client's optimism, or a politician may want to maximize voter engagement.

Receiver Learning and Sender Sampling The first result is that receiver learning is shaped by the interaction of the sender's modeling limitations with his sampling strategy. The sender provides a model, shown in purple in the figure below. However, the receiver may still entertain alternative models. These plausible models may have higher or lower γ values. Under local sampling, the receiver's set of plausible models is tangent to the sender's model at a single point. At x^* — where the sender's model is maximized — models with higher γ (orange) are below the receiver's model. Models with lower γ (green) are above the sender's model. Under distant sampling, the set of plausible models shifts vertically, shown on the right. Because all models share the same sample mean, higher γ models lie above the sender's model at x^* and lower γ ones lie below.



(a) Plausible models under local sampling

(b) Plausible models under distant sampling

Figure 1: Plausible models when the sender's model understates tradeoffs

This leads to a second result: when the sender's models tend to *overstate* tradeoffs, he benefits from local sampling. In this case, any additional models the receiver entertains have lower γ than the sender's. Concretely, the green lines in the figure lie above the sender's model at x^* under local sampling, but below them under distant sampling. The sender's perception of the receiver's expected utility is higher in the former case. However, if the sender's models *understate* tradeoffs, he prefers distant sampling. The receiver's additional models have higher γ than the sender, lying above the sender's model under distant sampling but below under local sampling. A key corollary is that a sender restricted to *linear regression* models ($\gamma = 0$) always prefers distant sampling. This raises a tension between the interpretation of a regression as a "local linear approximation" to a data-generating process and its strategic use by a sender to manipulate a receiver's optimism.

Third, this interaction between model misspecification and sampling has implications for receiver welfare. We extend the receiver's prfeferences to account for *forecast error*, in addition to maximizing the expected value of the outcome. Under distant sampling, the vertical dispersion across plausible models increases at x^*, so if the receiver values forecast accuracy, she tends to prefer local sampling. Thus, the receiver learns the true DGP more precisely when the sender overstates tradeoffs and samples *locally*.

Gains from Oversimplification Fourth, a sender may gain from model oversimplification. I characterize when a "sophisticated" sender — who can fit any γ in the receiver's feasible set — benefits from the presence of "simple" sender types restricted to a tighter range of γ values. Upon observing a model, the receiver believes she received the correct model from the sophisticated sender or a misspecified model from the simple sender. I show that the sophisticated sender gains when the simple sender samples in a way such that any alternative models lie above the reported model. Crucially, this relies on the flexibility that simple senders have in their ability to *sample* *information*, meaning they can disclose information in a way that credibly allows the sophisticated sender's model to be a "lower bound" for the receiver's expected utility.

Moreover, I show that if a "type-space designer" could choose an optimal distribution of simple sender types to maximize payoffs, he would have one sender for each γ value who can only fit models with that specific γ value. The key implication is that organizations benefit from deploying senders with a fixed—but arbitrary—way of modeling the tradeoff intensity γ.

Sender Competition The final set of results study competition between multiple senders who face overhead costs of providing a model. This section is in the process of being written up in the current draft of the paper. We model two senders who set prices and additionally receive a reputational gain from making R optimistic. We show that if the cost of model provision is relatively high, R only buys a model from a single sender who under/overstates tradeoffs. Thus, model simplicity continues to survive into settings with competition. However, the subtlety is that because senders receive a reputational gain from providing R a simple model, they may be willing to price their models below cost.

However, if the cost of model provision is relatively low and, in addition, the magnitude of the reputational gain from supplying a simple model is small, a receiver may be willing to purchase models from both senders instead of buying a single simple model from one. On the one hand, this generates perfect learning, since R can piece together the true model from both of their pieces of information. On the other hand, this erases senders' reputational gain from model simplification. Senders are no longer willing to price below cost, generating both an intensive margin (prices) and extensive margin (number of models) discontinuity in expenditures for the sender.

Section 2 lays out the basic model, and I characterize learning in section 3. Section 4 studies gains from oversimplifying models and the design of an optimal type space for a sophisticated sender. Section 5 investigates how these distortions become change when considering competition between multiple senders. Section 6 concludes. The remainder of this section gives an overview of the literature.

Literature This paper broadly adds to our understanding of information design and learning when utilizing *models of the world*. It particularly focuses on the interaction of *model simplicity* with *data sampling* techniques. Similar relations have been studied by Eliaz and Spiegler (2020) — who show that models that organize past data to make senders hopeful can generate self-sustaining political polarization; Montiel Olea et al. (2022) — who show that agents tasked with prediction adopt more and more complex models as they have access to more data; Schwartzstein and Sunderam (2021) — who study receiver learning when senders trade off model fit and belief manipulability; and Aina (2023) — who characterizes the limits of belief manipulability when a receiver provides a sender with multiple models for interpreting data. These phenomena have also been studied in the lab (Aina and Schneider, 2025; Barron and Fries, 2024).

The particular angle the present paper takes is related to omitted variable bias — in that a sender cannot (or refuses to) fully model a tradeoff in the way a receiver would like. This connects the paper to work like Levy et al. (2022), Gleyze and Pernoud (2022), and Eliaz et al. (2025) — who show that model misspecification may interact with competitive pressures or uncertainty about data estimation to generate a range of outcomes from improved communication and engagement to political cycles and the prevalence of false models. The second part of the present paper studies competition among senders and markets for models, which relates it to work such as Dasaratha et al. (2025).

Because of its reliance on models as information structures, the present paper also adds to a longer literature on (Bayesian) persuasion (Kamenica and Gentzkow, 2011), as well as persuasion and competition (Gentzkow and Kamenica, 2017; Wang, 2013). In particular, receivers in the present paper are uncertain as to the source of data or the broader class of models a sender's model was obtained from. This connects the paper to models of persuasion and cheap talk with *uncertainty* about information structures (Huang, 2024; Min, 2026) and ambiguity aversion (Colo, 2024). Conceptually, the insight that some senders' gains from providing simple models can be eroded by sequential information acquisition is similar to that in Matyskova and Montes (2023).

Many of the examples in the paper concern *delegation* to experts. There is long line of literature — particularly in organizations and finance — studying this process, including mistrust in experts (Cheng and Hsiaw, 2022), uncertainty about portfolio managers' models (Li and Wang, 2018), uncertainty about expertise (Kishishita, 2020), and its interaction with persuasion Kolotilin and Zapechelnyuk (2025).

A key finding of this paper is that senders may generally benefit from providing *simpler* models to receivers. The idea that senders can benefit from simple models is also captured by Perez-Richet and Prady (2012), Bloedel and Segal (2018), Lipnowski et al. (2020), Eliaz et al. (2021), and Burkovskaya and Starkov (2026). While many of these papers are driven by exogeneous restrictions on parameter spaces, bounded rationality, or rational inattention, the findings in this paper are related to *uncertainty*; a receiver cannot observe the sort of data the sender estimated the model on, or what other models the sender may have considered (which in turn interacts with senders' sampling strategies). Mathematically, receivers in the paper differ in their ability to *model tradeoffs*, which are related to their ability to estimate models on predetermined *subsets* of a parameter space. This characterization of commitment — as well as the problem of the "sender-type designer" — is thus conceptually related to Bizzotto et al. (2026)'s characterization of the limits of commitment in quantity-price games.

2 Benchmark Model

We begin by laying out a game between a single receiver R (she) and a single sender S (he). R decides whether she would like to contract the services of a sender S. S collects data and provides a R a model describing that data. Neither S's set of models nor the data sampling strategy are observed by R. R utilizes the information provided in S's model to make a decision. We abstract from prices and competition, but return to study these questions in Section 5.

2.1 Receiver and Models

A receiver R faces a set of choices $X = \mathbb{R}$. Each of these choices maps to a distribution of outcomes $y \in \mathbb{R}$. Ex-ante, R does not know the distribution of outcomes. However, she believes that the distribution of y given x can be described by a data-generating process $y = f_R(x)$ for a function $f_R : X \to \mathbb{R}$. We refer to f_R as a **model**. We denote $\mathcal{F}$ as the set of *models* f_R that R entertains. $\mathcal{F}$ is closed and convex. We make the following assumptions on $\mathcal{F}$.

Assumption 1. *For each $f_R \in \mathcal{F}$, we have*

$$f_R(x) = \alpha_0 + \alpha_1 x - \gamma x^2,$$

where $\alpha_0 \sim P_0^R$ on $\mathbb{R}$, $\alpha_1 \sim P_1^R$ on $\mathbb{R}$, and $\gamma \sim P_\gamma^R$ on $[0, \overline{\gamma}] := \Gamma$ for measures P_0^R, P_1^R, P_γ^R. Additionally, for all $\beta_0, \beta_0' \in \mathbb{R}$, $P_0^R(\{\beta_0\}) = P_0^R(\{\beta_0'\})$.

Assumption 1 says that each of R's functions is concave, and can in particular be expressed as an affine function less a convex, quadratic loss. Notably, since $-\gamma x^2$ is concave, the functional form can also represent the consideration of diminishing returns in consumer utility, production, or any other concave function in shaping an outcome. We assume a quadratic form for tractability but show that the central intuitions generalize in section 3.5. Moreover, the prior over each of the three components is independent. Finally, the prior over the set of constant terms is flat, i.e. R's true uncertainty is over the curvature of f_R.

We will use the notation q to refer to R's belief over the set of models, i.e. their (joint) belief over these three parameters. Given a belief q over models, R's expected utility is given by:

$$U_q(x) = \mathbb{E}_q[f_R(x)].$$

Her goal is to choose x to maximize $U_q(x)$. The receiver values the expected value of the outcome y given her choice x. If R is an investor, she cares about how much money she can make by investing in a choice x. If R is a CEO restructuring her firm via layoffs, she cares about her decision's impact on profits. If R is a voter, she cares about the impact of a policy x on her wellbeing. We utilize these simpler preferences in our baseline model, but section 3.4 extends them to evaluate settings where R also values *forecast accuracy*, and thus knowing the correct model of the world.

2.2 Sender and Data

The receiver R contracts a sender S to provide her with information about the true data-generating process. The sender has a private type affecting the simplicity of the models she considers. The order of events is as follows.

1. Nature draws S's type and the receiver's true model f_R.

2. S collects data.

3. S provides a model to R that fits the data he collected. R updates her beliefs about the true data-generating process based on her beliefs about S's type and the model.

4. R makes a decision x^*. S receives utility. Uncertainty resolves, and R receives her utility.

We now provide more detail about each of these components.

Sender's Type A sender's type affects the set of models — and hence information — he is capable of providing to R. The key is that S is limited in how he can model the non-affine component of f_R. To this end, we denote the type space by $\Theta \subseteq \mathcal{P}(\Gamma)$ — i.e. each type $\theta \in \Theta$ corresponds to a subset of $[0, \overline{\gamma}]$. The sender's type is drawn from a distribution T_θ over Θ, which is known to R. Θ has the following properties.

Assumption 2. *For each $\theta \in \Theta$, $\theta \subseteq \Gamma$ is a (potentially degenerate) closed interval. Moreover, for each $\theta \neq \theta'$, $\theta \cap \theta' = \varnothing$.*

Assumption 2 states that each type corresponds to an interval within $[0, \overline{\gamma}]$. This interval may be degenerate. We assume the intervals do not intersect for expositional purposes, but the case where they do not intersect is of particular interest, and we study it in section 4.

A type θ sender is able to provide R a model in the set F_θ, where

$$F_\theta := \{f_R \in \mathcal{F} : f_R = \beta_0 + \beta_1 x - \delta x^2, \delta \in \theta, (\beta_0, \beta_1) \in \mathbb{R}^2\}.$$

Intuitively, θ represents the *degree* to which a sender can model the *non-linear* component of a data-generating-process f_R. For example, suppose there is a single type $\theta = \Gamma$. Then, S is able to provide R any of the models in $\mathcal{F}_R$. Of particular interest is the case where $\theta = \{0\}$, in which case S can only provide information to R in the form of a "linear regression." For ease of notation, we will denote $\underline{\delta}_\theta = \min\{\theta\}$ and $\overline{\delta}_\theta = \max\{\theta\}$

We assume a type θ sender has a prior over $\alpha_0, \alpha_1, \delta$ given by measures $\alpha_0 \sim P_0^\theta$ and $\alpha_1 \sim P_1^\theta$ with support on $\mathbb{R}$ and $\delta \sim P_\gamma^\theta$ with support on $\theta \subseteq \Gamma$.

Sender's Data Prior to choosing a model, S chooses a data sampling strategy μ_θ, where μ_θ is a distribution defined over X. Intuitively, this means that S is choosing a distribution of choices in x to collect data on. R does not observe S's choice of μ_θ, meaning data-collection suffers from a *lack of verifiability and commitment*.

Assumption 3. *μ_θ is parametrized by $\mu_\theta(x_\theta, \sigma) = x_\theta + \sigma \cdot \epsilon$, where ϵ is mean-zero, symmetric, and* $\mathrm{Var}(\epsilon) = 1$. *$x_\theta$ takes a value in a compact, nondegenerate interval $[\underline{x}, \overline{x}]$ Moreover, σ takes a value in a compact, nondegenerate interval $[\underline{\sigma}, \overline{\sigma}]$ for $\underline{\sigma} > 0$ arbitrarily small.*

Assumption 3 provides a convenient parametrization of μ_θ as a distribution centered around a mean x_θ. σ controls the standard deviation of the distribution. The constraint that $\sigma > 0$ is a normality assumption made to ensure learning is structured in equilibrium. x_θ and σ_θ will denote the equilibrium choices of a type θ sender.

Given this sampling strategy, S directly observes the joint distribution of (y, x), subject to observational noise. Formally, if the true data-generating-process is f_R^*, she observes a joint distribution $D(y, x)$, where the marginal distribution of x is given by $D_x = \mu_\theta$ and the conditional distirbution of y given x is given by $D_y(y|x) = f_R^*(x) + \eta$, where η is a mean-zero noise term with unknown variance. This is equivalent to S observing an infinite number of i.i.d. draws from the true data-generating-process.

Sender's Information With the sender's type and sampling strategy, in hand, we define the model S provides to R. We say that a model f_S **best-fits the data** with respect to θ and a data-sampling strategy μ_θ if:

$$f_S = \arg\min_{f \in F^\theta} \mathbb{E}_{D(\mu_\theta)}\left[\left(y - f(x)\right)^2\right].$$

The existence of a best-fitting model is given by the Hilbert Projection Theorem. For example, if $F^\theta = \mathcal{F}$, then a best-fitting f_S will represent the true data-generating process. Or, if $F^\theta = \mathcal{L}$, f_S will be the best-fitting *linear regression model* to the data. Our key assumption will be that the model f_S provides must best-fit the data, given his set of models F^θ.

Sender's Preferences Suppose S provides a model $f_S \in \mathcal{F}_\theta$. Let x^* be the receiver's choice of $x \in X$ conditional on her posterior beliefs over models $q|f_S$. The sender's utility conditional on f_S is given by:

$$U_S(x^*) = \mathbb{E}_{q|f_S}[f_R(x^*)].$$

That is, conditional on providing a model f_S, S would like to maximize R's expected utility. These preferences lend themselves to several interpretations. For example:

- S is a portfolio manager who receives a fixed share of R's expected earnings. Then, he cares about the expected profits R can realize (and thus his payment).
- S is a consulting firm whose services a CEO R contracts. S may care about the expected gain in profits R receives either directly (e.g. because his "bonus" is proportional to R's gain in profits) or indirectly (e.g. because his reputation is tied to his ability to realize profits for those who contract his services).
- S is an office-motivated politician who provides information to a median voter R. The higher the median voter thinks y is, the more likely she is to turn out. If re-elected, the politician will implement R's preferred policy x^*.

2.3 Comments

The model assumes that R cannot verify information on two fronts: *data* and *functional form*. The lack of commitment to a specific dataset underscores a key feature of contracting with models. R knows S has access to useful data, but does not know exactly what that data looks like or how it is being used to provide her with a model. The clearest example is one where S's data is exclusive and proprietary — R cannot observe these data directly, and must take into account S's incentives and the information she is provided to make inferences about what the data might be. Portfolio managers and consulting firms may have access to past data on returns or cases they have worked on which they are unable to disclose directly, or politicians may have access to confidential data on past policy experiments collected by the public service. Another example is an AI model which detects high-level data patterns in a "black-boxed"' way that cannot be elucidated to a user (or even its designers).

More broadly, however, S's data may not literally be "data," but rely on curation of existing experience and institutional knowledge. A consultant, for example, may have experience working on various cases, meaning the expertise he is bringing to R's problem also embeds her working knowledge of R's situation. Additionally, R may also be unable (or unwilling) to expend organizational time and resources in collecting, understanding, and synthesizing complex data. Instead, she delegates the collection of data and its synthesis into a model to the sender, expediting her decision-making process in the interest of convenience, but potentially at the cost of information.

This is related to the second dimension of nonverifiability, which is that S may not necessarily fit the true data-generating-process in the way R would like. The sender's type constrains how he can model the true data-generating-process, which may be affected by technological feasibility or a difference in thinking about a problem the same way R does. For example, suppose S is a climate scientist and R is an economist. S provides climate modeling to help R assess the impact of a climate shock on the demand for electric vehicles. Even if S understands the concept of demand, there may be technical constraints on the sorts of parameters his advanced climate models can provide to R. In particular, because $-\gamma x^2$ represents a convex loss, model simplicity particularly constrains S's ability to model *tradeoffs*. Thus, S may be forced to provide a model that understates or overstates the degree of a tradeoff.

Our key assumption is that S must provide the best-fitting model to the data, rather than a model that simply "fits the data." The interpretation is that S makes an honest effort to provide R the best model he can, given the set of data-generating-processes he can model. While it could be that S's honesty emerges due to professional norms or unmodeled career concerns, it may also be that S synthesizes data using methods or algorithms that are well-known in his field of expertise. For example, an economist may always use linear regression models to fit the data; while he can select what variables to include, he nevertheless is always providing a model of best fit *within* a class of models. Relatedly, R could require S's methodology to be "replicable." Nevertheless, even though S provides a model of best-fit, he can still manipulate the set of models R entertains through the way he initially samples data as well as his type θ.

Many of the results will involve thinking about what happens when S can only provide linear regression models to R. This is motivated by the widespread usage of linear regression models in the business world, economics, and data-driven social sciences. We will be interested in understanding not only how we can learn about non-linear processes through linear models, but also how the use of linear models coupled with data sampling strategies can be used to manipulate beliefs about the effects of a policy or intervention. Moreover, we will contrast how the behavior of senders constrained to provide linear models may in particular differ from the behavior of senders constrained to provide other simplified (but nonlinear) models.

For the majority of the analysis, we focus on settings where the nonlinear component of f_R is quadratic. While this is done for tractability, section 3.4 discusses the sorts of assumptions necessary to generalize the logic to more general functional forms. Most of the results also generalize to multidimensional settings, and many of the proofs are identical, which we discuss in the same section.

3 Learning from Models

3.1 Receiver's Learning and Behavior

We begin by characterizing receiver learning as a function of data-sampling strategies and S's type.

Receiver's Updating Suppose S provided a model of the form $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$ which best fits the data. First, note that due to Assumption 2, R is able to infer θ from δ. We define $\phi(f_S; \theta, \mu_\theta)$ as the set of functions that could rationalize the data for R, given f_S best-fits with respect to F_S and μ_θ:

$$\phi(f_S; \theta, \mu_\theta) = \left\{ f_R \in \mathcal{F} : f_S = \arg\min_{f \in F^\theta} \mathbb{E}_{\mu_\theta}\big[\big(f_R(x) - f(x)\big)^2\big] \right\}.$$

Crucially, ϕ describes the set of DGPs that R entertains which S is unable to model but R believes are plausible. Since S's type does not constrain β_0, β_1, we necessarily have that for each $f_R \in \phi(f_S; \theta, \mu_\theta)$:

$$\frac{\partial}{\partial \beta_i} \mathbb{E}_{\mu_\theta}\big[\big(f_R(x) - f(x)\big)^2\big] = 0,$$

which yields the following moment conditions familiar from ordinary-least-squares (OLS):

$$\mathbb{E}_{\mu_\theta}\big[f_R(x) - \beta_0 - \beta_1 x + \delta x^2\big] = 0 \quad (1)$$

$$\mathbb{E}_{\mu_\theta}\big[x\big(f_R(x) - \beta_0 - \beta_1 x + \delta x^2\big)\big] = 0 \quad (2)$$

Equation (1) is the moment condition with respect to the intercept, and states that on average, the difference between f_R and f_S must average to zero. We note that this is the same as the constraint that a model must "fit the data." Equation (2) can be combined with equation (1) to read that the covariance between x and $f_R(x) - \beta_0 - \beta_1 x - \delta x^2$ is zero. Both expectations are taken with respect to μ_θ, which is S's sampling strategy.

Additionally, we also have a moment condition corresponding to δ:

$$\mathbb{E}_{\mu_\theta}\big[x^2\big(f_R(x) - \beta_0 - \beta_1 x + \delta x^2\big)\big] \tag{3}$$

which may be zero or nonzero, depending on S's type.

The following proposition characterizes the set of functions that R entertains given f_S, θ, and μ_θ.

Proposition 1. *Suppose $f_S = \beta_0 + \beta_1 x - \delta x^2$ best-fits the data and $\mu_\theta = \mu_\theta(x_\theta, \sigma)$. Then, there exists a unique $\theta \ni \delta$ such that for all $f_R \in \phi(f_S; \theta, \mu_\theta)$, we have:*

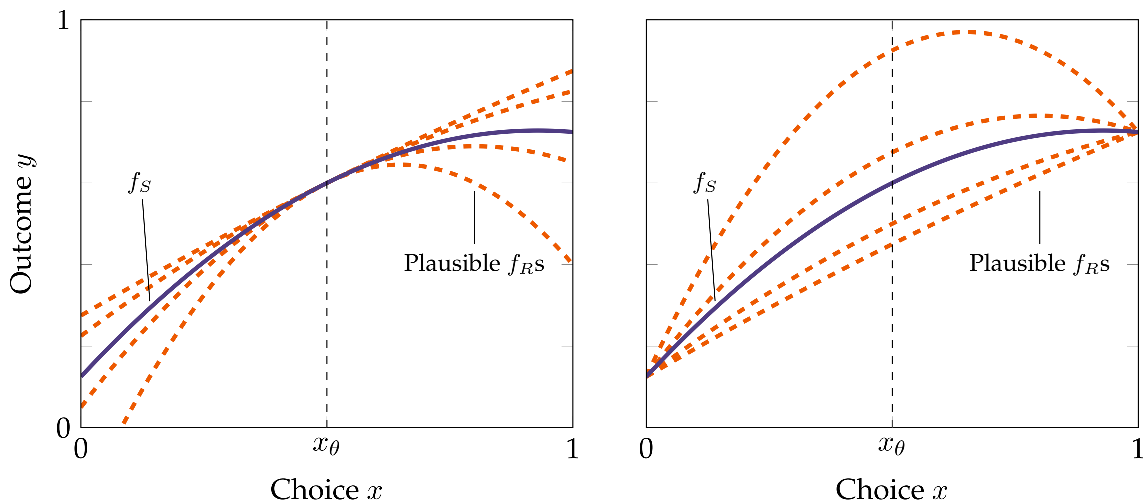
$$f_R(x) = f_S(x) + (\gamma - \delta)\big(\sigma^2 - (x - x_\theta)^2\big).$$

The set of models that R considers, given a best-fitting model f_S, takes a simple mathematical form: they are all the sender's original model plus a constant, less a quadratic term centered at x_θ. Both of these terms are scaled by $\gamma - \delta$. Geometrically, these curves are all vertically-fanned out versions of the sender's original model with the same slope at x_θ but different degress of concavity.

Next, at x_θ, the derivative of both f_S and f_R with respect to x is $\beta_1 - 2\delta$. Thus, all plausible f_Rs have the same slope at x_θ. Finally, each of these models may have a potentially different quadratic component γ than the sender's original model. Notably, if $\theta = \Gamma$, ϕ will be a singleton containing the true DGP. Otherwise, there may be multiple models that rationalize the data.

The intuition can be described as follows. First, we utilize a result known as "Stein's Lemma" that the derivative of f_R and f_S must be the same "on average" over μ_θ. As $\sigma \to 0$, for each $f_R \in \phi(f_S; \theta, \mu_\theta)$, the derivative of f_R and f_S converges at the point x_θ. Using (1), we also have that as $\sigma \to 0$, the value of each f_R and f_S must converge at f_S. Thus, all $f_R \in \phi(f_S; \theta, \mu_\theta)$ are (approximately) tangent to f_S at x_θ.

This intuition is shown in the left panel of Figure 2 below. The solid line shows f_S, which has a curvature δ. However, given the sender's type θ may not account for all potential values that γ could take, there are other models f_R which could have generated the data, which are in turn reflected in the dashed blue lines. Crucially, all these lines are tangent to f_S at x_θ.



(a) Plausible models when $\sigma = \underline{\sigma}$

(b) Plausible models when $\sigma > 0$

Figure 2: Plausible receiver models f_R as function of sender model f_S and sampling strategy $\mu_\theta(x_\theta, \sigma)$

The set of feasible models is different for $\sigma > 0$. In particular, if $\sigma' > \sigma$, then the distribution $\mu_\theta(x_\theta, \sigma')$ is a mean-preserving spread of $\mu_\theta(x_\theta, \sigma)$. Since f_R and f_S are both quadratic, their derivatives are linear, meaning (2) is invariant to mean preserving spreads (and thus does not depend on σ). Thus, each plausible f_R at σ is also plausible at σ' and vice-versa, up to a constant. The value of this constant depends on the curvature of f_R relative to f_S. Because each f_R is concave, its expectation over $\mu_\theta(x_\theta, \sigma)$ is decreasing in σ. If a given f_R is "more concave" than f_S — i.e. if $\gamma > \delta$, this f_R must shift up by a constant relative to f_S. If a given f_R is "less concave" than f_S — i.e. if $\gamma < \delta$, f_R must shift down by a constant relative to f_S. This is shown in the right panel of Figure 2. For higher σ, the plausible f_Rs that are "more linear" lie below f_S in the sampling range. The f_Rs that are "more concave" lie above f_S.

Receiver's Decisionmaking The receiver's learning takes a relatively simple form: regardless of the limitation on S's ability, R always learns what the data-generating-process looks like "up to the first derivative." Moreover, as σ changes — and thus S's sampling strategy changes — the set of plausible f_Rs remains identical, except that they shift up or down by a constant. This means that R's optimal *action* depends only her expectation of γ (given δ and θ). Given f_S, the expected value of the problem also has a simple expression.

Corollary 1. *R's optimal action is given by:*

$$x^* = x_\theta + \frac{\beta_1 - 2\delta x_\theta}{2\mathbb{E}^R_\gamma[\gamma|\theta, \delta]}.$$

R's expected value from this action is given by

$$y^* = \beta_0 + \beta_1 x_\theta - \delta x_\theta^2 + \frac{(\beta_1 - 2\delta x_\theta)^2}{4\mathbb{E}_\gamma^R[\gamma|\theta,\delta]} + \big(\mathbb{E}_\gamma^R[\gamma|\theta,\delta] - \delta\big)\sigma^2.$$

The value of $\mathbb{E}_\gamma^R[\gamma|\theta,\delta]$ is also relatively simple.

- If (3) is equal to 0 at δ, then $\mathbb{E}_\gamma^R[\gamma|\theta,\delta] = \delta$.
- If (3) is ≤ 0, then it must be that $\delta = \overline{\delta}_\theta$, and $\mathbb{E}_\gamma^R[\gamma|\theta,\delta] = \mathbb{E}[\gamma|\gamma \geq \delta]$.
- Likewise, if (3) is ≥ 0, then it must be that $\delta = \underline{\delta}$, and $\mathbb{E}_\gamma^R[\gamma|\theta,\delta] = \mathbb{E}_\gamma^R[\gamma|\gamma \leq \delta]$.

3.2 Sender's Data Sampling and Model Provision

We use these insights to derive the first result of the paper, which shows how S chooses a sampling strategy (i.e. x_θ and σ_θ) as a function of her type. We will use the following terminology to distinguish different types of sampling.

- **Local sampling**: $\sigma = \underline{\sigma}$. In this case, μ_θ samples x values in a small neighborhood of x_θ, i.e. is collecting information "locally relevant" to x_θ.
- **Distant sampling**: $\sigma = \overline{\sigma}$. In this case, μ_θ samples x values from as wide a range as possible x_θ.

We make the following assumption on P_γ^θ.

Assumption 4. *Suppose $\theta = [\underline{\delta}_\theta, \overline{\delta}_\theta]$ is non-degenerate. Then,*

- *if $\underline{\delta}_\theta > 0$, $P_\gamma^\theta(\{\underline{\delta}_\theta\}) > 0$ (i.e. P_γ^θ has a mass point at $\underline{\delta}_\theta$).*
- *If $\overline{\delta}_\theta < \overline{\gamma}$, $P_\gamma^\theta(\{\overline{\delta}_\theta\}) > 0$ (i.e. P_γ^θ has a mass point at $\overline{\delta}_\theta$).*

We impose Assumption 4 as a regularity condition; it is consistent with either P_γ being a discrete distribution, or P_γ being a continuous CDF over $[0, \overline{\gamma}]$ whose mass is artificially truncated on $[\underline{\delta}_\theta, \overline{\delta}_\theta]$. Next, we introduce the following notation

$$\begin{aligned}\overline{\gamma}^\theta &= \mathbb{E}_\gamma^R[\gamma|\gamma \geq \overline{\delta}_\theta] \\ \underline{\gamma}^\theta &= \mathbb{E}_\gamma^R[\gamma|\gamma \leq \underline{\delta}_\theta]\end{aligned}$$

These represent R's expectation of γ, given γ potentially takes values that S cannot model.

Finally, we introduce the following definitions. We say a type θ sender **understates (overstates) tradeoffs** if either:

- $\theta^\circ \neq \varnothing$ and $P_\gamma^\theta(\{\overline{\delta}_\theta\})(\overline{\gamma}^\theta - \overline{\delta}_\theta) > (<) P_\gamma^\theta(\{\underline{\delta}_\theta\})(\underline{\delta}_\theta - \underline{\gamma}^\theta)$;

- or $\theta = \{\delta_\theta\}$ and $\mathbb{E}^R_\gamma[\gamma] < (>) \delta_\theta$.

Concretely, if a sender *understates* tradeoffs, he systematically tends to "underestimate" the true magnitude of the quadratic term relative to the receiver. In the first case, where θ is nondegenerate, the sender's prior may tend to place more mass on $\overline{\delta}_\theta$ than $\underline{\delta}_\theta$, there may be a large gap between $\overline{\delta}_\theta$ and the receiver's expectation of γ given it is higher than $\overline{\delta}_\theta$, or a combination. In the second case where θ is degenerate — meaning the sender can only model a singleton δ value — that δ value is lower than the receiver's prior expectation. A receiver *overstates* tradeoffs if the opposite happens: he tends to "overestimate" the true magnitude of the quadratic term relative to the receiver.

We now state our result on sender sampling.

Theorem 1. *If S understates tradeoffs, he distantly samples ($\sigma_\theta = \overline{\sigma}$) and sets $x_\theta = \max\{\underline{x}, \min\{\frac{\mathbb{E}^\theta_{\beta_1}[\beta_1]}{2\mathbb{E}^\theta_\gamma[\gamma]}, \overline{x}\}\}$. If S overstates tradeoffs, he locally samples ($\sigma_\theta = \underline{\sigma}$) and sets $x_\theta = \underline{x}$ or $\overline{x}$.*

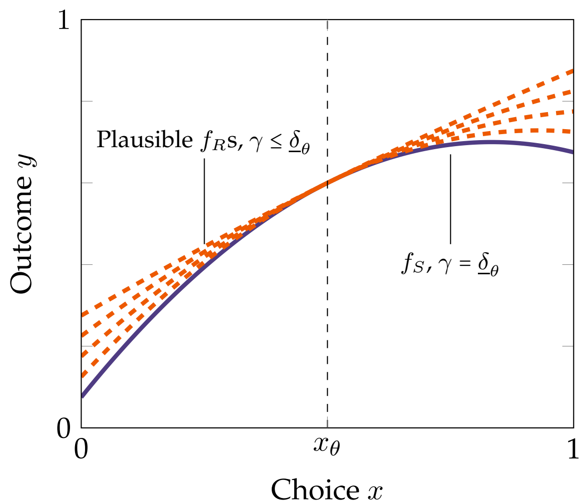
Theorem 1 can be interpreted as follows. If $\delta \in (\underline{\delta}_\theta, \overline{\delta}_\theta)$, then R believes S's models is correct. However, if $\delta = \overline{\delta}_\theta$, R will consider models with $\gamma \geq \overline{\delta}_\theta$. In this case, S's tends to estimate models that understate the magnitude of a potential tradeoff. This creates an incentive for him to estimate his model on information that is potentially irrelevant to learning about the DGP in a neighborhood of x_θ.

However, when $\delta = \underline{\delta}_\theta$, the opposite happens: R considers models with $\gamma \leq \underline{\delta}_\theta$, i.e. "flatter" than those that S provided. This gives S an incentive to estimate the DGP "locally" in a neighborhood of x_θ. If a type θ S tends to consider models with only high values of γ — i.e. to "overstate" the sharpness of a tradeoff — then he does have an incentive to locally sample choices in a neighborhood of x_θ.

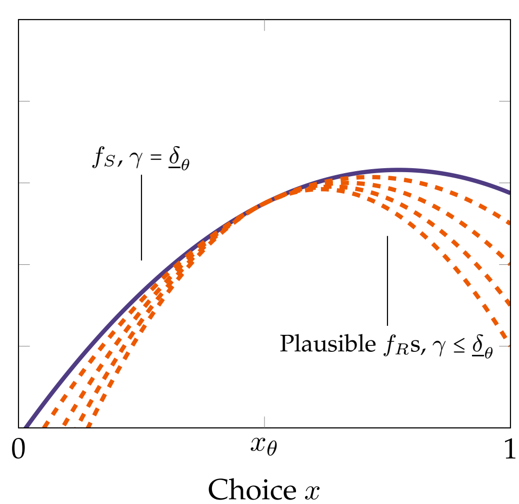
The intuition is most easily seen in Figure 3 below. The left panels (a) and (c) look at the case where f_S is such that $\delta = \underline{\delta}_\theta$, and thus where R entertains models that are "less steep" than f_S. The right panels (b) and (d) look at the case where f_S is such that $\delta = \overline{\delta}_\theta$, and R entertains steeper models (with higher γ).

In panels (a) and (b), we first look at the case where $\sigma = \underline{\sigma}$ and S locally samples around x_θ. In both cases, the set of plausible models f_R is approximately tangent to f_S at x_θ. However, in panel (a), because these f_Rs are "shallower" (i.e. have lower γ values than $\underline{\delta}_\theta$), each function lies above f_S. By contrast, in panel (b), each f_R is "steeper" than f_S (i.e. has $\gamma \geq \overline{\delta}_\theta$), and thus lie *below* f_S.

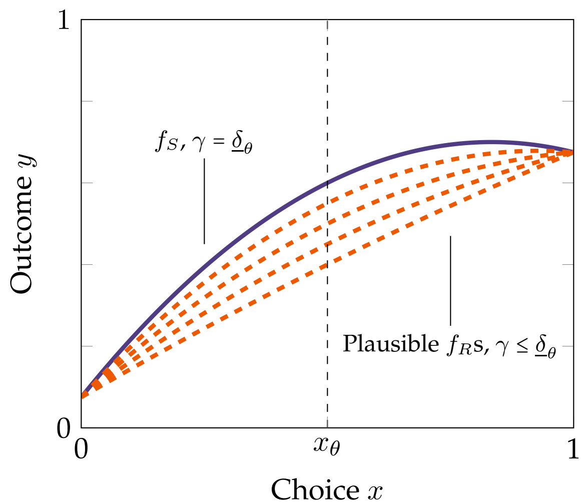
This logic flips when looking at panels (c) and (d). Recall that the distribution of x values when $\sigma = \overline{\sigma}$ is a mean-preserving-spread of that when $\sigma = \underline{\sigma}$. To rationalize the data, functions with higher γ values that are "more concave" must shift up relative to shallower ones. In panel (c), this means that the shallower f_R functions now lie *below* f_S. In panel (d), the steeper f_R functions now lie *above* f_S.



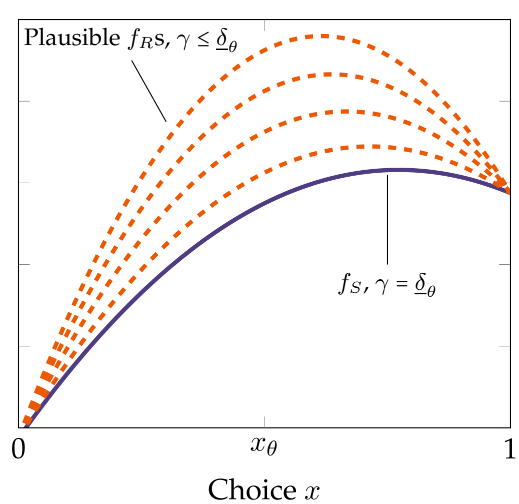
(a) Plausible f_Rs when $\sigma = \underline{\sigma}$, $\delta = \underline{\delta}_\theta$



(b) Plausible f_Rs when $\sigma = \underline{\sigma}$, $\delta = \overline{\delta}_\theta$



(c) Plausible f_Rs when $\sigma = \overline{\sigma}$, $\delta = \underline{\delta}_\theta$



(d) Plausible f_Rs when $\sigma = \overline{\sigma}$, $\delta = \overline{\delta}_\theta$

Figure 3: Models R entertains when $\delta = \underline{\delta}_\theta$ vs. $\delta = \overline{\delta}$, local sampling ($\sigma = \underline{\sigma}$) vs. distant sampling ($\sigma = \overline{\sigma}$)

Thus, whether $\sigma_\theta = \overline{\sigma}$ or $\underline{\sigma}$ depends on which scenario is more likely for S. If he is likely to end up in situations where the set of functions R will see is "shallower" than the "shallowest" function his type can model, he will prefer local sampling, as this effectively makes his model a lower bound for the set of models R entertains. If he is likely to end up in situations where the set of functions R will see is "steeper," he will prefer distant sampling.

In an extreme case, suppose $\theta = [\underline{\delta}_\theta, \overline{\gamma}]$. In this case, S can model the steepest tradeoffs R will entertain, but not shallower ones. Thus, he will always prefer local sampling. On the other hand, suppose $\theta = [0, \overline{\delta}_\theta]$. Then, any additional models that R entertains are "steeper" than f_S, so he has an incentive to distantly sample data and drive up R's expected value of these steeper problems.

Finally, we provide intuition for the value of x_θ. Under tradeoff understatement and distant sampling, S would like to ensure all the additional models that R entertains are as far above f_S^θ as possible. This gap is largest when x_θ is at the maximum of f_S^θ. Because S does not know ex-ante where this maximum will be, he thus samples where he *expects* f_S^θ to be the shallowest.

Under tradeoff overstatement and local sampling the logic is the opposite: if f_S^θ is maximized at a point x^*, so are all other models tangent to f_S^θ. However, if all these other models are tangent at a point farther from x^*, the set of tangent models grow higher and higher above f_S^θ. Thus, he samples at either $\underline{x}$ or $\overline{x}$, where the set of models S obtains are likely to be the steepest.

Of particular interest is the situation where $\theta = \{0\}$, which corresponds to a type of S who can only model *linear regressions*. Using Theorem 1, not only does such a sender always have an incentive to sample distantly, but as $\sigma \to \infty$, his perception of R's utility from his model also goes to ∞.

Corollary 2. *Suppose $\theta = \{0\}$, so that S only sends linear models. Then, S distantly samples. As $\sigma \to +\infty$, $V(x_\theta) \to +\infty$. S's perceptions about R's value also go to $+\infty$.*

This logic is easily seen in the figure below: Fixing the sender's (linear regression) model, as σ increases, the set of models the receiver entertains shift higher and higher. While linear regression models are highly tractable to estimate, their simplicity is often justified as based on "local linear approximations" of nonlinear data-generating processes. However, a sender who could *provide* a linear model would instead want to include data from all across the policy space.

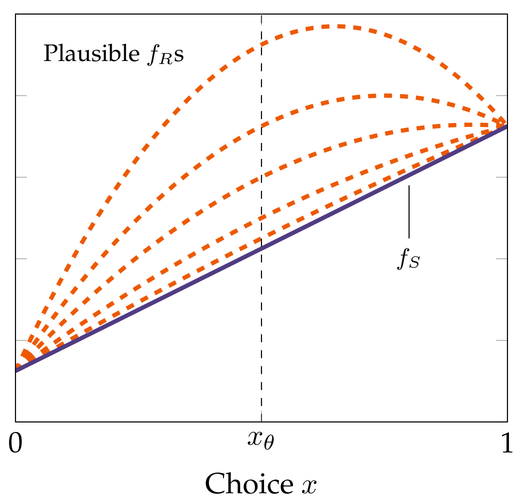


Figure 4: Plausible receiver models f_R when f_S is a linear regression

3.3 Receiver's Value of Information

In the absence of the sender, R would solely rely on her prior to choose x^*. This allows us to characterize the *value of information* for the sender.

Proposition 2. *The value of information for the receiver is given by*

$$V^I(T,\Theta) = \frac{1}{4}\left(\int_\Theta \mathbb{E}^R_{\alpha_1,\gamma}\big[\alpha_1^2 q(\gamma,\theta)\big] dT(\theta) - \frac{\mathbb{E}^R_{\alpha_1,\gamma}[\alpha_1]^2}{\mathbb{E}^R_{\alpha_1,\gamma}[\gamma]} \right)$$

where

$$q(\gamma,\theta) = \begin{cases} \frac{1}{\underline{\gamma}^\theta} & \gamma \le \underline{\delta}_\theta \\ \frac{1}{\gamma} & \gamma \in (\underline{\delta}_\theta, \overline{\delta}_\theta) \\ \frac{1}{\overline{\gamma}^\theta} & \gamma \ge \overline{\delta}_\theta \end{cases}$$

if $\underline{\delta}_\theta < \overline{\delta}_\theta$ *and* $q(\gamma,\theta) = \frac{1}{\mathbb{E}^R_\gamma[\gamma]}$ *otherwise.*

Suppose a type θ sender entertains precisely the same models as the receiver — i.e. is such that $0 = \underline{\delta}_\theta < \overline{\delta}_\theta = \overline{\gamma}$. In this case, the value of information is proportional to $\mathbb{E}^R_{\alpha_1,\gamma}\big[\frac{\alpha_1^2}{\gamma}\big] - \frac{\mathbb{E}^R_{\alpha_1}[\alpha_1]^2}{\mathbb{E}^R_\gamma[\gamma]}$. In this case, the receiver gains (relative to the prior) from both not only the local slope at x_θ (β_1) but also the steepness of the curvature γ. As $\underline{\delta}_\theta$ increases or $\overline{\delta}_\theta$ decreases, the receiver learns less and less (in expectation) about γ. Finally, when $\underline{\delta}_\theta = \overline{\delta}_\theta$, the receiver effectively learns nothing about the curvature. However, she still retains value from hiring the sender, since she learns about the slope of f_R. In this final case, the value of information is proportional to $\frac{\mathrm{Var}^R_{\alpha_1}(\alpha_1)}{\mathbb{E}^R_\gamma[\gamma]}$.

Crucially, in the case of a receiver who maximizes $\mathbb{E}[y|x]$, the receiver's *expected utility* does not depend on the sender's choice of σ. The intuition is that different sampling strategies cause upwards and downwards shifts in the set of potential models the receiver entertains, which have no bearing on the *choice* the receiver eventually makes. Distant and local sampling will, however, affect the *variance* of the receiver's outcome, which we turn to next.

3.4 Forecast Accuracy

So far, we have assumed only that the receiver would like to maximize $\mathbb{E}[y|x]$. However, in many settings, the receiver may also directly (or indirectly) value *forecast accuracy*. A CEO, for instance, may not only value making the *right decision* about workforce restructuring, but may benefit from *accurately forecasting* the results of her restructuring. In fact, the less she is able to defend her choice with the "correct model," the more backlash she may face from the board of directors or employees at her firm. A receiver who values *forecast accuracy* intuitively prefers *model certainty*, which we have abstracted from so far.

To illustrate the impact of model simplicity and sampling on forecast-minded receivers, we consider a receiver whose expected utility is given by

$$(1-\psi)\mathbb{E}^R[y|x] - \psi \mathrm{Var}^R(y|x)$$

for $\psi \in (0,1)$. Intuitively, these preferences can also be thought of as follows. In a first period, the receiver forms an expectation of y. In a second period, given this expectation, the utility she reaps is based on her ability to forecast. ψ measures her patience; if $\psi = 0$, she does not value the future or forecasting, and only cares about her expectation of y. As $\psi \to 1$, she places a greater and greater emphasis on forecasting. We write a forecast-minded receiver's value as

$$V^F(f_S|\theta,\sigma) = \max_x (1-\psi)\mathbb{E}_{f_S|\theta,\sigma}[y|x] - \psi \mathrm{Var}_{f_S|\theta,\sigma}[y|x].$$

Proposition 3. *Fix $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$. Then, the following results holds.*

1. *If x^* maximizes $\mathbb{E}^R_{f_S|\theta,\sigma}[y|x]$, then $\mathrm{Var}^R_{f_S|\theta,\sigma}[y|x^*]$ is increasing in σ for σ sufficiently large.*
2. *$V^F(f_S|\theta,\sigma)$ is decreasing in σ for σ sufficiently large.*
3. *If $f'_S(x_\theta) = 0$, $V^F(f_S|\theta,\sigma)$ is everywhere decreasing in σ.*
4. *If $f'_S(x_\theta) \neq 0$, $V^F(f_S|\theta,\sigma)$ is increasing in σ for σ small.*

The intuition for the first part of the result is as follows. As σ increases, the set of models the receiver entertains shift upward or downward. While the receiver's actual choice of x does not change, the expected value of y given x spreads out. Thus, $\mathrm{Var}^R_{f_S;\theta,\sigma}[y|x]$ is increasing in σ^2.

This means that, if the receiver were simply to choose x^* — which would maximize $\mathbb{E}[y|x]$, distant sampling would be *worse* for the receiver, in the sense that her choices would be *less accurate*. Combined with Theorem 1, this suggests that a receiver is better off when facing a sender whose models tend to *overstate* tradeoffs — who have an incentive to distantly sample — than those whose models tend to *understate* tradeoffs — who *locally* sample.

The first part in Proposition 3 considers the case where the receiver does not internalize the variance of y when choosing x^*. However, the next three parts consider what happens when R makes her choice of x^* *internalizing* the variance of outcomes.

The key insight in this case is that the variance of y is precisely minimized when the receiver's models cross the sender's at the points $x_\theta \pm \sigma$. This is illustrated in Figure 5. Thus, R trades off between a choice $x = x^\dagger$ that maximizes her expected utility (at an interior point) and minimizing the variance of outcomes, which happens at $x_\theta \pm \sigma$. When σ is very large, R is forced to choose arbitrarily extreme points $x_\theta \pm \sigma$ to minimize variance. This decreases expected utility sufficiently, giving the second result: V^F is decreasing in σ for σ large.

When σ is small, the logic is more subtle. If $f'_S(x_\theta) = 0$, then her expected utility is simply maximized at x_θ itself: both f_S and all other models R considers achieve a maximum at x_θ. Howeve, for any $\sigma > 0$, there is always a tradeoff between setting $x = x_\theta$ and $x = x_\theta \pm \sigma$. Thus, when $f'_S(x_\theta) = 0$, R's utility is everywhere decreasing in σ. This is the third result.

However, if $f'_S(x_\theta) \neq 0$, R can be made better off by increasing σ. In particular, R is best off when $\sigma = x^\dagger - x_\theta$. In this case, choosing $x^\dagger$ both maximizes the expected value of y and ensures the expected variance of outcomes is minimized. This gives the final result: if $f'_S(x_\theta) \neq 0$, R's utility is increasing in σ for σ small.

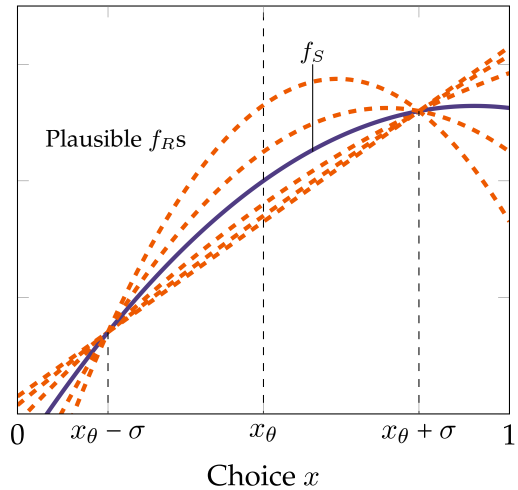


Figure 5: Intersection of receiver's models with sender's models

Thus, even if the receiver places vanishingly small weight on $\mathbb{E}[y|x]$ and values forecasting error, the sender still has an incentive to depart from local sampling. Moreover, unlike in the baseline case, there may be an incentive for the sender to sample more distantly if the receiver values forecast error.

3.5 Generalized Model

While the main model utilizes quadratic utility to express tradeoffs, in this section, we probe robustness of many of the main results to *non-quadratic* utility. In particular, instead of a quadratic loss $-\gamma x^2$, we allow for loss to take the form $-\gamma c(x)$, where $c(x)$ is a convex loss function. In particular, we show an analogue of Theorem 1: if $\overline{\sigma}$ is sufficiently large and S tends to understate tradeoffs, distant sampling dominates local sampling; and if S tends to overstate tradeoffs, local sampling dominates distant sampling.

This section is under construction! Please click the link to see if this is available in a future draft. Argument outline:

- Consider convex loss function $\gamma \cdot c(\cdot)$
- Fixing γ, establish equivalency class: $\beta_0 + \beta_1 x - \gamma \cdot c(\cdot) \cong \alpha_0 + \alpha_1 x - \gamma \cdot c(\cdot)$. These have the same weight in the prior.
- Under mean preserving spreads: if "load" change in moment condition (2) onto β_1 or α_1, show that value of all functions increases at x_θ. So things "shift up," although slope at x_θ might go up or down.
- Theorem: if $\overline{\sigma}$ is sufficiently large, sampling with $\overline{\sigma}$ dominates $\underline{\sigma}$.

4 Gains from Simplification

So far, we have taken model oversimplification as an exogenous feature of our informational environment. This section studies when a sender benefits from simplification, thus providing an *endogenous* channel for the sender to provide simplified models.

In the baseline model, we assumed that senders' types (as subintervals of $[0, \overline{\gamma}]$) were exogeneous and non-overlapping. We first return to this setting but allow senders' types to overlap. We show that when there are different types of senders, they benefit only when a sender's type intersects with the *boundary* of another sender's type, and the probability of any boundary intersection has nonzero measure. However, if one sender is made better off by having an overlying type, the other is necessarily made worse off, thus providing a channel for senders' types to not overlap in equilibrium.

Using these intuitions, we then consider the problem of an "omniscient" type space designer who has a prior over δ on the entirety of $[0, \overline{\gamma}]$. This designer can either choose to provide the true model, or to draw different types of senders with different probabilities. The problem can be thought of as akin to a consulting firm "drawing" teams of consultants that can only think about tradeoffs in specific ways, or an AI model designer "drawing" different models that are artifically constrained in how they can think.

We show that if this designer considers has a continuous prior over γ, the optimal design of the type space is composed of *singletons*. In particular, for every $\gamma \in [0, \overline{\gamma}]$, there is a type $\theta = \{\gamma\}$. Thus, sender types each have a fixed (but arbitrary) of modeling the receiver's tradeoff, meaning the receiver *never* learns the true value of γ. This result emerges both due to the *form* of model simplicity (uncertainty about the magnitude of tradeoffs) and different senders' use of strategic sampling.

We also characterize the solution where the designer has a discrete prior over γ, in which case the optimal design involves either *singleton* γ values or pooling at most *two* γ values into a single type.

(The case where the designer has a continuous prior with mass points does not have a well designed solution. But, under a perturbation assumption, the optimal type space is composed of an interval and singleton partition, where the upper bound of each interval occurs on a mass point, and "interval" senders always distantly sample. We return to characterize this solution in a future draft of the paper.)

4.1 Characterizing Benefits from Oversimplification

Previously, we assumed that senders differed in their ability to model the quadratic component of the receiver's models, that different senders could model this component in distinct ways, and thus that the receiver could fully back out the sender's type from the model she received. However, we now relax Assumption 2 and allow the sender's types to have overlapping supports. I.e., we consider the case where there exist $\theta, \theta' \in \Theta$ such that $\theta' \cap \theta \neq \varnothing$.

Let $\delta \in \theta \cap \theta'$. Conditional on seeing a model of the form $y = \beta_0 + \beta_1 x - \delta x^2$, the receiver does not know whether this model came from a type θ or type θ' sender. In particular, we can characterize when uncertainty about the source of the signal can actually work in a sender's benefit, and when it hurts the sender.

Proposition 4. *Consider two types of senders $\theta = [\underline{\delta}, \overline{\delta}]$ and $[\underline{\delta}', \overline{\delta}']$ such that $\theta \cap \theta' \neq \varnothing$ and $\theta \cap \theta' \subseteq \theta^\circ$. Suppose R receives information from either sender with positive probability. Fix a sender model $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$ such that $\delta \in \theta \cap \theta'$. Given $f_S(x)$, a type θ sender's utility strictly increases (decreases) from the existence of type θ' if one of the following hold.*

1. *$\delta \in \theta^\circ \neq \varnothing$, $\delta = \overline{\delta}'$, and θ' distantly (locally) samples.*
2. *$\delta \in \theta^\circ \neq \varnothing$, $\delta = \underline{\delta}'$, and θ' locally (distantly) samples.*
3. *$\delta = \overline{\delta}$, $\theta^\circ = \varnothing$, type θ locally samples, and θ' distantly (locally) samples.*
4. *$\delta = \underline{\delta}$, $\theta^\circ = \varnothing$, type θ distantly samples, and θ' locally (distantly) samples.*
5. *$\theta^\circ = \varnothing$, $\theta'^\circ \neq \varnothing$, $\delta = \overline{\delta}'$, and θ' distantly (locally) samples.*
6. *$\theta^\circ = \varnothing$, $\theta'^\circ \neq \varnothing$, $\delta = \underline{\delta}'$, and θ' locally (distantly) samples.*

The first part of the proposition shows that the receiver's *maximal expected utility* tends to be higher from models with higher γ than the sender's under distant sampling. Although the proposition has many cases, they can be grouped as follows.

The first two cases correspond to situations where the sender's model has a δ value in the *interior* of θ. Without θ', R would simply deduce that this was the true model. However, with θ', R may entertain *higher* γ models or *lower* γ models if δ is on the boundary of δ'. If $\delta = \overline{\delta}'$ and θ' distantly samples, these lead to a gain in S's expected utility, since any higher γ models lie *above* the original model. If $\delta = \underline{\delta}'$ and θ' locally samples, R entertains lower γ models that lie *below* the original model.

The next two cases consider situations where δ is at the boundary of θ. It may be that case that $\delta = \overline{\delta}$, but that a type θ sender ex-ante thought $\underline{\delta}$ was more likely and thus locally sampled, even though ex-post he would have preferred distant sampling. In this case, he is made better by a sender who considers the same models but distantly samples. There is an analogous case for $\delta = \underline{\delta}$.

The final two cases consider situations where $\theta = \{\delta\}$ is a singleton. In this case, if R thinks the model came from type θ, she will consider both higher and lower γ value models. However, if $\delta = \overline{\delta}'$ and θ'°, she may think that model came from θ'. Because $\theta^\circ \neq \varnothing$, she will *only* consider models with higher γ that lie *above* the original model. A similar logic applies to $\delta = \underline{\delta}'$.

A corollary of this result, however, is that if both θ and θ' senders emerge with positive probability, interactions are zero-sum: if one sender is made better off by another, the other is made worse off.

Corollary 3. *Conditional on providing a model $f_S(x)$, suppose a type θ sender's expected utility increases from a type θ' sender's existence. Then, the type θ' sender's expected utility decreases from type θ sender's existence.*

4.2 Optimal Type Space Design

Using these results, we consider the problem of a "sophisticated" type-space designer D. D has a prior over β_0, β_1, δ given by P_0^D, P_1^D, P_γ^D. We write as P_{f_S} his prior over models f_S.

We suppose that D faces the following problem. Conditional on a model f_S probability q_0, R believes she receives the true model (and does not consider alternative models). With probability $1 - q_0$, she believes the true model comes from the type space Θ. The leading interpretation is that R contracts from multiple senders, and always deals with a "truthful" sender who provides the true model with an exogenous probability.

The order of events is as follows. D first chooses a type space Θ such that all elements in Θ are closed intervals in Γ (potentially degenerate). Then, D chooses a distribution over types T_θ. The sender contracts the services of a sender, and receives a model from the "true" sender with probability q_0 or from the type space Θ with probability $1 - q_0$. In the latter case, a sender is randomly drawn from the type space Θ according to T_θ. In either case, the deployed sender supplies the model of best fit.

The optimal characterization of the type space is as follows.

Theorem 2. *Suppose P_γ^D is continuous. Then, the optimal type space is the collection of singletons on $[0, \overline{\gamma}]$: $\Theta = \{\{\delta\} : \delta \in \Gamma\}$. Moreover, T_θ places positive mass on all points with positive mass in P_γ^R. D's utility is increasing in q_0.*

This result relies on many of the intuitions of Proposition 4. Consider a model of the form $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$. For each δ, we introduce a sender of type $\{\delta\}$. We refer to this sender as an "arbitrary" type sender, in the sense that he provides models with an arbitrarily fixed value of δ. Conditional on seeing $f_S(x)$, R would believe that it either came from the "true" type of sender or the "arbitrary" type of sender. In the former case, the receiver would believe f_S is the true model. In the latter case, because the arbitrary type of sender sample strategically, R's maximal expected utility from considering alternative models would be higher than from just consider f_S. The more likely the latter case, the higher R's maximal expected utility is, from the designer's perspective; thus D't utility is increasing in q_0. The designer thus always gains from creating an "arbitrary" sender for each $\delta \in \Gamma$ and matching the probabilities of each of these types to the value of γ he entertains in his prior.

Naturally, D could also introduce a sender of type $\theta = [\underline{\delta}^\theta, \overline{\delta}]$ with $\theta^\circ \neq \varnothing$. In this case, the assumption that P_γ^D has a continuous density is crucial. In particular, this sender could distantly sample, meaning that conditional on seeing $\delta = \overline{\delta}$, R would entertain even higher expected utility models than if she were to think it only came from an arbitrary type $\{\overline{\delta}\}$ sender. To guarantee this even sharper gain from model simplification, for any model with $\delta \in \theta^\circ$, type θ would necessarily have to provide the *true model*, erasing any potential gains from tradeoff under/overstatement. Since $\overline{\delta}$ is a singleton, it has measure 0 in D's prior, while θ° has measure greater than 0.

Because D's utility is increasing in q_0, if possible, D would send $q_0 \to 1$. In this sense, a type space where senders use a fixed (but entirely arbitrary) way of modeling tradeoffs can be seen as the limiting case for a situation where a designer begins by modeling the "true model" and then artificially simplifying his model.

As a result, we also consider separately the case where P^D_γ is *discrete*. In this case, the optimal type space consists of both *singletons* and *doubles* — i.e. D either provides an arbitrary model or pools together no more than two values of δ together. We write $\Gamma = \{\gamma_i\}_{i \in \mathbb{N}}$.

Proposition 5. *Suppose the type space designer's prior P^D_γ over Γ is discrete. Then the optimal type space is given by a partition of Γ into a combination of singletons $\{\delta_i\}$ and doubles $\{\delta_i, \delta_{i+1}\}$.*

Version 2: The order of events is as follows. D first chooses a type space $\Theta \subseteq \mathcal{P}(\Gamma)$. Θ must have the following properties.

1. For each $\theta \in \Theta$, θ must be a closed (potentially degenerate) interval.

2. If $\theta^\circ \neq \varnothing$, there exists Δ arbitrarily small such that

The first property is definitional. The second ensures the existence of a well-defined solution to the problem.

The designer chooses a distribution of T_θ over Θ. The receiver then contracts a sender to investigate a choice x_θ. A type θ sender is randomly drawn from Θ according to T_θ. That sender strategically chooses a sampling strategy μ_θ and provides a model f^θ_S to the receiver who then updates.

D's expected utility is the same as senders': he would like to maximize R's expectation of her utility, given she sees a model f^θ_S and knows type θ sender samples using a distribution μ_θ. From Proposition 3 and Theorem 1, we can expression R's expected utility, conditional on seeing a model $f_S = \beta_0 + \beta_1 - \delta x^2$ as

$$V^R(f_S) = \max_x \mathbb{E}_{\theta|f_S}\Big[f_S(x) + (\mathbb{E}^R[\gamma|\delta,\theta] - \delta)\big(\sigma^2_\theta - (x - x_\theta)^2\big)\Big].$$

In turn, D in turn integrates V^R over his prior over models

$$U(T_\Theta, \Theta) = \int_\Theta \int_{\mathcal{F}_R} V^R(f_S) dP(f_S|\theta) dT(\theta) \tag{4}$$

Theorem 3. *The optimal type space design is characterized by an interval $[\underline{\delta}, \overline{\delta}]$. If $\underline{\delta} > 0$, there is a type $\underline{\theta} = [\underline{\delta}, \underline{\delta} + \Delta$. If $\overline{\delta} < \overline{\gamma}]$, there is a type $\overline{\theta} = [\overline{\delta} - \Delta, \overline{\delta}]$. Then, Θ is a collection of singletons with two intervals at the end: Then, $\Theta \underline{\theta} \cup \overline{\theta} \cup \{\delta\}_{\delta \in [\underline{\delta}, \overline{\delta}] - \left(\overline{\theta} \cup \underline{\theta}\right)}$. If $[\underline{\delta}, \overline{\delta}] = [0, \overline{\gamma}]$, then Θ is a collection of singletons: $\Theta = \{\{\delta\} : \delta \in \Gamma\}$.*

5 Competition

This section is largely under construction! See a sketch below, but please click the link to see if this is available in a future draft.

We now consider competition between multiple senders who provide information at market prices. Following the previous section, we consider senders drawn from arbitrary type spaces, i.e. such that $\Theta = \{\{\delta\} : \delta \in \Gamma\}$. In particular, we consider two senders.

We assume that senders each incur a cost of c for providing models. This could represent the price of collecting data for the sender, as well as the labor required to fit the model, present the model to the receiver, and other managerial or overhead expenses. The two senders compete over prices. R purchases a model from whomever provides his services at a lower price. This is due to the fact that R cannot distinguish the two senders; thus, even if one sender precommitted to always providing the true model (instead of a simplified one),

A type θ sender's profits from receiving a contract from R are given by price less cost, plus a reputational component. Profits are written as follows.

$$p - c + \omega \mathbb{E}_{f_R}\left(\mathbb{E}_{f_S^\theta}\left[\max_x \mathbb{E}_{f_R | f_S^\theta}[f_R(x)] - \max_x f_R^\theta(x) \right] \right) \qquad \omega > 0 \tag{5}$$

The main innovation is the reputational component multiplied by $\omega > 0$. The interpretation is that R values not only immediate profits from the project, but also a further gain from making S optimistic (such as follow on contracts, a gain from the approval of a project, or additional reputational considerations). The key difference in this case is that the reputational gain is normalized to 0 relative to R's true model.

Proposition 6. *Suppose R only buys from one sender. Then, both senders set $p < c$.*

The intuition is as follows. Suppose both senders priced at c. In this case, the reputational gain for both senders would be positive. However, because each type θ samples stragecically, R entertains models that lie above f_S^θ, meaning the reputational gain is positive. Thus, both senders are willing to price below c.

The key insight, however, is that this requires R to purchase information from *only* a single sender. However, R could in fact purchase models from both senders simultaneously. In this case, the result can generate an increase in prices. Recall in this case that the value of information from the "arbitrary" type space is given by

$$V^I = \frac{1}{4} \frac{\mathrm{Var}_{\alpha_1}^R(\alpha_1)}{\mathbb{E}_\gamma^R(\gamma)}$$

Write the equilibrium of this model as $c - \omega\kappa$. By contrast, the value from full information is given by

$$V^F = \frac{1}{4}\mathbb{E}^R_{\alpha_1}\left(\frac{\alpha_1^2}{\gamma}\right) - \frac{\mathbb{E}^R_{\alpha_1}(\alpha_1)}{4\mathbb{E}^R_{\gamma}(\gamma)}$$

Then, we have the following result.

Theorem 4. *Suppose $V^F - 2c < V^I - c + \omega\kappa$. Then, $p = c$, and R buys from both senders in equilibrium.*

First, note that when R buys from both receivers, she is able to recover the true model; in particular, she has four conditions with which to identify the true model. In the case where R buys from both senders, this thus erases any of the gains ωk from making R optimistic, generating an increase in prices relative to the situation in which R only purchased from one sender.

For this to occur in equilibrium, it must be incentive compatible for R to want to purchase information from two senders. In particular, R must prefer to buy models from two senders as opposed to just buying from one (at a price below c). A necessary condition is that c is sufficiently low, but even then, it must be that the reputational gain (and hence the amount by which senders are willing to mark down prices) is also sufficiently small to justify not simply purchasing from one sender, even if the quality of information is lower.

6 Conclusion

This section is under construction! Please click the link to see if this is available in a future draft.

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A Proofs of Results

Proposition 1. *Suppose $f_S = \beta_0 + \beta_1 x - \delta x^2$ best-fits the data and $\mu_\theta = \mu_\theta(x_\theta, \sigma)$. Then, there exists a unique $\theta \ni \delta$ such that for all $f_R \in \phi(f_S; \theta, \mu_\theta)$, we have:*

$$f_R(x) = f_S(x) + (\gamma - \delta)\big(\sigma^2 - (x - x_\theta)^2\big).$$

Proof. We rewrite the sender's model as a local expansion around x_θ: $f_S(x) = \tilde{\beta}_0 + \tilde{\beta}_1(x - x_\theta) - \delta(x - x_\theta)^2$, where $\tilde{\beta}_1 = \beta_1 - 2\delta x_\theta$. Because sender type intervals θ are non-overlapping, the value of δ uniquely pins down the sender's type θ.

Suppose $f_R \in \phi(f_S; \theta, \mu_\theta)$. Similarly expanding f_R around x_θ, we have:

$$f_R(x) = \tilde{\alpha}_0 + \tilde{\alpha}_1(x - x_\theta) - \gamma(x - x_\theta)^2,$$

where k is a constant. Because μ_θ is symmetric, it is elliptical (in one dimension). By Stein's Lemma (Landsman and Nešlehová, 2008), equation (2) can be written as

$$\begin{aligned}
\mathbb{E}_{\mu_\theta}\big[x\big(f_R(x) - f_S(x)\big)\big] &= \mathrm{Cov}_{\mu_\theta}(x, f_R(x) - f_S(x)) \\
&= \sigma^2 \mathbb{E}_{\mu_\theta}[f_R'(x) - f_S'(x)] \\
&= \sigma^2 \mathbb{E}_{\mu_\theta}[\tilde{\alpha}_1 - 2\gamma(x - x_\theta) - \tilde{\beta}_1 + 2\delta(x - x_\theta)] = 0
\end{aligned}$$

Because the mean of μ_θ is x_θ, this yields the condition $\tilde{\alpha}_1 = \tilde{\beta}_1$ which is *independent* of σ^2. This means that $\alpha_1 - 2\gamma x_\theta = \beta_1 - 2\delta x_\theta$, i.e. $\alpha_1 = \beta_1 - 2(\delta - \gamma)x_\theta$ Hence, we have

$$f_R(x) = \alpha_0 + (\beta_1 - 2(\delta - \gamma)x_\theta)x - \gamma x^2,$$

with the value of γ and α_0 potentially indeterminate. However, fixing a plausible γ value, from (1) we have

$$\begin{aligned}
\mathbb{E}_{\mu_\theta}[f_R(x)] = \alpha_0 + (\beta_1 - 2(\delta - \gamma)x_\theta)x_\theta - \gamma(x_\theta^2 + \sigma^2) = \beta_0 + \beta_1 x_\theta - \delta(x_\theta^2 + \sigma^2) &= \mathbb{E}_{\mu_\theta}[f_S(x)] \\
\implies \alpha_0 - 2(\delta - \gamma)x_\theta^2 &= \beta_0 - \delta x_\theta^2 - \delta\sigma^2 \\
\implies \alpha_0 &= \beta_0 + (\gamma - \delta)(\sigma^2 - x_\theta^2)
\end{aligned}$$

Putting the pieces together, we have

$$\begin{aligned}
f_R(x) &= \beta_0 + (\gamma - \delta)(\sigma^2 - x_\theta^2) + (\beta_1 - 2(\delta - \gamma)x_\theta)x - \gamma x^2 \\
&= \beta_0 + \beta_1 x - \delta x^2 + \big(-(\gamma - \delta)x^2 + (\gamma - \delta)(\sigma^2 - x_\theta^2) + 2(\gamma - \delta)x_\theta\big) \\
&= \beta_0 + \beta_1 x - \delta x^2 + (\gamma - \delta)\big(\sigma^2 - (x - x_\theta)^2\big) \\
&= f_S(x) + (\gamma - \delta)\big(\sigma^2 - (x - x_\theta)^2\big)
\end{aligned}$$

□

Theorem 1. *If S understates tradeoffs, he distantly samples ($\sigma_\theta = \overline{\sigma}$) and sets $x_\theta = \max\{\underline{x}, \min\{\frac{\mathbb{E}^\theta_{\beta_1}[\beta_1]}{2\mathbb{E}^\theta_\gamma[\gamma]}, \overline{x}\}\}$. If S overstates tradeoffs, he locally samples ($\sigma_\theta = \underline{\sigma}$) and sets $x_\theta = \underline{x}$ or $\overline{x}$.*

Proof. Following from the Proposition 1, let $\theta = [\underline{\delta}_\theta, \overline{\delta}_\theta]$. Suppose $\theta^\circ \neq \varnothing$ and $\delta \in \theta^\circ$. Then (3) is necessarily satisfied, and $\phi(f_S; \theta, \mu_\theta) = \{f_S\}$. However, if $\delta = \underline{\delta}_\theta$ or $\overline{\delta}_\theta$, then (3) is a corner solution from the receiver's perspective. If $\delta = \overline{\delta}_\theta$, then any $\gamma \geq \overline{\delta}_\theta$ is plausible from the receiver's perspective. If $\delta = \underline{\delta}_\theta$, any $\gamma \leq \underline{\delta}_\theta$ is plausible. Finally, if $\underline{\delta}_\theta = \overline{\delta}_\theta$, any $\gamma \in [0, \overline{\gamma}]$ value is possible. This pins down the plausible set of γ values from the receiver's perspective.

Next, fix σ. The receiver chooses x to maximize:

$$\beta_0 + \beta_1 x - \delta x^2 + \big(\mathbb{E}^R_\gamma[\gamma|\delta] - \delta\big)\big(\sigma^2 - (x - x_\theta)^2\big) \tag{6}$$

The receiver's choice is given by $x^* = x_\theta + \frac{\beta_1 - 2\delta x_\theta}{2\mathbb{E}^R_\gamma[\gamma|\delta]}$. Write $x^* = x_\theta + h$. In this case, her expected value is:

$$\begin{aligned}
\beta_0 + \beta_1 x^* - \delta(x^*)^2 + \big(\mathbb{E}^R_\gamma[\gamma|\delta] - \delta\big)\big(\sigma^2 - (x^* - x_\theta)^2\big) \\
\beta_0 + \beta_1(x_\theta + h) - \delta(x_\theta^2 + 2x_\theta h + h^2) + \big(\mathbb{E}^R_\gamma[\gamma|\delta] - \delta\big)\sigma^2 - \mathbb{E}^R_\gamma[\gamma|\delta]h^2 + \delta h^2 \\
\beta_0 + \beta_1(x_\theta + h) - \delta(x_\theta^2 + 2x_\theta h) + \big(\mathbb{E}^R_\gamma[\gamma|\delta] - \delta\big)\sigma^2 - \mathbb{E}^R_\gamma[\gamma|\delta]h^2 \\
\beta_0 + \beta_1 x_\theta - \delta x_\theta^2 + (\beta_1 - 2\delta x_\theta)h + \big(\mathbb{E}^R_\gamma[\gamma|\delta] - \delta\big)\sigma^2 - \mathbb{E}^R_\gamma[\gamma|\delta]h^2 \\
\beta_0 + \beta_1 x_\theta - \delta x_\theta^2 + \frac{(\beta_1 - 2\delta x_\theta)^2}{2\mathbb{E}^R_\gamma[\gamma|\delta]} - \frac{(\beta_1 - 2\delta x_\theta)^2}{4\mathbb{E}^R_\gamma[\gamma|\delta]} + \big(\mathbb{E}^R_\gamma[\gamma|\delta] - \delta\big)\sigma^2 \\
\beta_0 + \beta_1 x_\theta - \delta x_\theta^2 + \frac{(\beta_1 - 2\delta x_\theta)^2}{4\mathbb{E}^R_\gamma[\gamma|\delta]} + \big(\mathbb{E}^R_\gamma[\gamma|\delta] - \delta\big)\sigma^2
\end{aligned} \tag{7}$$

Suppose $\delta \in \theta^\circ$. Then, $\mathbb{E}^R_\gamma[\gamma|\delta] = \delta$, and the receiver's expected value does not depend on σ^2. If $\delta = \overline{\delta}_\theta$, her expected utility is

$$\beta_0 + \beta_1 x_\theta - \delta x_\theta^2 + \frac{(\beta_1 - 2\delta x_\theta)^2}{4\overline{\gamma}^\theta} + (\overline{\gamma}^\theta - \delta)\sigma^2,$$

recalling that $\overline{\gamma}^\theta = \mathbb{E}^R_\gamma[\gamma|\gamma \geq \overline{\delta}_\theta]$. Similarly, we have that if $\delta = \underline{\delta}_\theta$, her expected utility is

$$\beta_0 + \beta_1 x_\theta - \delta x_\theta^2 + \frac{(\beta_1 - 2\delta x_\theta)^2}{4\underline{\gamma}^\theta} + (\underline{\gamma}^\theta - \delta)\sigma^2.$$

Working backwards, S then chooses σ^2 to maximize

$$P^\theta_\gamma(\{\overline{\delta}_\theta\})\Bigg(\frac{\mathbb{E}^R_\gamma[(\beta_1 - 2\delta x_\theta)^2]}{4\overline{\gamma}^\theta} + (\overline{\gamma}^\theta - \overline{\delta}_\theta)\sigma^2\Bigg) + P^\theta_\gamma(\{\underline{\delta}_\theta\})\Bigg(\frac{\mathbb{E}^R_\gamma[(\beta_1 - 2\delta x_\theta)^2]}{4\underline{\gamma}^\theta} + (\underline{\gamma}^\theta - \underline{\delta}_\theta)\sigma^2\Bigg).$$

The derivative with respect to σ^2 is simply

$$P_\gamma^\theta(\{\overline{\delta}_\theta\})(\overline{\gamma}^\theta - \overline{\delta}_\theta) - P_\gamma^\theta(\{\underline{\delta}_\theta\})(\underline{\delta}_\theta - \underline{\gamma}^\theta).$$

If the first term is greater than the second, the optimal choice is $\sigma = \overline{\sigma}$. Otherwise, $\sigma = \underline{\sigma}$. Finally, in the case where θ is degenerate and equal to $\{\delta\}$, the preference for distant sampling reduces to $\mathbb{E}_\gamma^R[\gamma] - \delta \geq 0$. This gives the result for sampling.

Finally, we characterize the equilibrium value of x_θ, $x_|theta$. Returning to (6) and using the envelope theorem, we have that the derivative with respect to x_θ is:

$$2\big(\mathbb{E}_\gamma^R[\gamma|\delta] - \delta\big)(x^* - x_\theta) \propto (\mathbb{E}_\gamma^R[\gamma|\delta] - \delta)\frac{\beta_1 - 2\delta x_\theta}{\mathbb{E}_\gamma^R[\gamma|\delta]}.$$

This is (strictly) concave if $\mathbb{E}_\gamma^R[\gamma|\delta] - \delta > 0$, and achieves a maximum when $\beta_1 = 2\delta x_\theta$. Given S must commit to R beforehand and integrating backwards, this gives $x_\theta = \frac{\mathbb{E}_{\beta_1}^\theta[\beta_1]}{2\mathbb{E}_\gamma^\theta[\gamma]}$ under tradeoff understatement. Otherwise, if $\mathbb{E}_\gamma^R[\gamma|\delta] - \delta \leq 0$, the function is convex and achieves a maximum at a corner solution $\underline{x}$ or $\overline{x}$, giving the potential values for x_θ.

□

Proposition 2. *The value of information for the receiver is given by*

$$V^I(T,\Theta) = \frac{1}{4}\Bigg(\int_\Theta \mathbb{E}_{\alpha_1,\gamma}^R\big[\alpha_1^2 q(\gamma,\theta)\big]dT(\theta) - \frac{\mathbb{E}_{\alpha_1,\gamma}^R[\alpha_1]^2}{\mathbb{E}_{\alpha_1,\gamma}^R[\gamma]}\Bigg)$$

where

$$q(\gamma,\theta) = \begin{cases} \frac{1}{\underline{\gamma}^\theta} & \gamma \leq \underline{\delta}_\theta \\ \frac{1}{\gamma} & \gamma \in (\underline{\delta}_\theta, \overline{\delta}_\theta) \\ \frac{1}{\overline{\gamma}^\theta} & \gamma \geq \overline{\delta}_\theta \end{cases}$$

if $\underline{\delta}_\theta < \overline{\delta}_\theta$ *and* $q(\gamma,\theta) = \frac{1}{\mathbb{E}_\gamma^R[\gamma]}$ *otherwise.*

Proof. Recall that given a sender model $f_S^\theta(x) = \beta_0 + \beta_1 x - \delta x^2$, each of R's plausible models can be written as:

$$f_R(x) = f_S^\theta(x) + (\gamma - \delta)\big(\sigma_\theta^2 - (x - x_\theta)^2\big),$$

where there is residual uncertainty about the value of γ. Throught this proof, with some abuse of notation, we use $\mathbb{E}$ to refer to R's expectations.

First, in the absence of information, R chooses x^* to maximize

$$\mathbb{E}[\alpha_0] + \mathbb{E}[\alpha_1]x - \mathbb{E}[\gamma]x^2 \implies x^* = \frac{\mathbb{E}[\alpha_1]}{2\mathbb{E}[\gamma]}$$

which gives an expected value of $\mathbb{E}[\alpha_0] + \frac{\mathbb{E}[\alpha_1]^2}{4\mathbb{E}[\gamma]}$.

First, suppose $\theta = [\underline{\delta}_\theta, \overline{\delta}_\theta]$ is nondegenerate. If $\delta \in \theta^\circ$, then $f_R(x) = f_S^\theta(x)$ and the value of information is

$$\mathbb{E}\left[\frac{\alpha_1^2}{4\gamma}\right] - \frac{\mathbb{E}[\alpha_1]^2}{4\mathbb{E}[\gamma]}.$$

However, suppose there is uncertainty about the value of γ. From the (7), R's conditional expected value is:

$$\beta_0 + \beta_1 x_\theta - \delta x_\theta^2 + \frac{(\beta_1 - 2\delta x_\theta)^2}{4\mathbb{E}[\gamma|\delta]} + \big(\mathbb{E}[\gamma|\delta] - \delta\big)\sigma^2.$$

Recalling that $\beta_1 - 2\delta x_\theta = \alpha_1 - 2\gamma x_\theta$, this can be rewritten as

$$\begin{aligned}
&\beta_0 + \big(\mathbb{E}[\gamma|\delta] - \delta\big)\sigma^2 + \big(\alpha_1 - 2(\mathbb{E}[\gamma|\delta] - \delta)x_\theta\big)x_\theta - \delta x_\theta^2 + \frac{(\alpha_1 - 2\mathbb{E}[\gamma|\delta]x_\theta)^2}{4\mathbb{E}[\gamma|\delta]} \\
&= \beta_0 + \big(\mathbb{E}[\gamma|\delta] - \delta\big)\sigma^2 + \alpha_1 x_\theta - 2\mathbb{E}[\gamma|\delta]x_\theta^2 + \delta x_\theta^2 + \frac{\alpha_1^2}{4\mathbb{E}[\gamma|\delta]} - \alpha_1 x_\theta + \mathbb{E}[\gamma|\delta]x_\theta^2 \\
&= \beta_0 + \big(\mathbb{E}[\gamma|\delta] - \delta\big)\sigma^2 + (\delta - \mathbb{E}[\gamma|\delta])x_\theta^2 + \frac{\alpha_1^2}{4\mathbb{E}[\gamma|\delta]}. \\
&= \overbrace{\beta_0 + \big(\mathbb{E}[\gamma|\delta] - \delta\big)(\sigma^2 - x_\theta^2)}^{\text{Noisy signal about } \alpha_0} + \frac{\alpha_1^2}{4\mathbb{E}[\gamma|\delta]}.
\end{aligned}$$

Note that because the first term is a constant and R is risk neutral in y, it does not affect the value of information for her. Thus, we are only left to consider the latter two terms. The non-α_0 set of terms take the following values:

$$\frac{\alpha_1^2}{4\gamma} \text{ for } \gamma \in (\underline{\delta}_\theta, \overline{\delta}_\theta). \qquad \frac{\alpha_1^2}{4\overline{\gamma}^\theta} \text{ for } \gamma \geq \overline{\delta}_\theta, \qquad \frac{\alpha_1^2}{4\underline{\gamma}^\theta} \text{ for } \gamma \leq \underline{\delta}_\theta.$$

Defining $q(\gamma, \theta)$ as in the statement of the proposition, integrating over α_1, γ, and then integrating again over the type space Θ gives the result. □

Proposition 3. *Fix $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$. Then, the following results holds.*

1. *If x^* maximizes $\mathbb{E}^R_{f_S|\theta,\sigma}[y|x]$, then $\mathrm{Var}^R_{f_S|\theta,\sigma}[y|x^*]$ is increasing in σ for σ sufficiently large.*
2. *$V^F(f_S|\theta,\sigma)$ is decreasing in σ for σ sufficiently large.*
3. *If $f_S'(x_\theta) = 0$, $V^F(f_S|\theta,\sigma)$ is everywhere decreasing in σ.*
4. *If $f_S'(x_\theta) \neq 0$, $V^F(f_S|\theta,\sigma)$ is increasing in σ for σ small.*

Proof. With some abuse of notation, let x^* maximize $\mathbb{E}[y|x]$ for R. Given the sender's model, the receiver's only uncertainty is over γ. This means that

$$\mathrm{Var}(y|x^*) = \begin{cases} 0 & \gamma = \delta \in (\underline{\delta}_\theta, \overline{\delta}_\theta) \\ \mathrm{Var}_{\gamma \geq \overline{\delta}_\theta}\big((\frac{\alpha_1^2}{4\gamma} + (\gamma - \overline{\delta}_\theta)(\sigma^2 - x_\theta^2)\big) & \gamma \geq \overline{\delta}_\theta \\ \mathrm{Var}_{\gamma \leq \underline{\delta}_\theta}\big((\frac{\alpha_1^2}{4\gamma} + (\gamma - \overline{\delta}_\theta)(\sigma^2 - x_\theta^2)\big) & \gamma \leq \underline{\delta}_\theta \end{cases}$$

In the case where $\gamma \geq \overline{\delta}_\theta$, this can be simplified as

$$\frac{\alpha_1^4}{16}\mathrm{Var}\Big(\frac{1}{\gamma}\Big) + (\sigma^2 - x_\theta^2)^2\mathrm{Var}(\gamma),$$

which is increasing in σ as long as $\sigma^2 \geq x_\theta^2$ A similar condition holds for the case where $\gamma \leq \underline{\delta}_\theta$. This gives the first part of the result.

We now turn to the second and third part of the results. Recall that:

$$f_R(x) = f_S(x) + (\gamma - \delta)(\sigma^2 - (x - x_\theta)^2) \implies \mathrm{Var}(y|x) = \big(\sigma^2 - (x - x_\theta)^2\big)^2\mathrm{Var}(\gamma|\delta).$$

Using this, R's objective can be written as

$$V^F(f_S|\theta,\sigma) = \max_x (1-\psi)\big(f_S(x) + (\mathbb{E}[\gamma|\delta] - \delta)(\sigma^2 - (x - x_\theta)^2)\big) - \psi\big(\sigma^2 - (x - x_\theta)^2\big)^2\mathrm{Var}(\gamma|\delta) \quad (8)$$

As above, the $1-\psi$ terms corresponding to $\mathbb{E}[y|x]$ are maximized at $x^\dagger = x_\theta + \frac{\beta_1 - 2\delta x_\theta}{2\mathbb{E}[\gamma|\delta]}$. The ψ term corresponding to $\mathrm{Var}(y|x)$ is minimized when $x = x_\theta \pm \sigma$. Increase σ large enough so that $x^\dagger \in (x_\theta - \sigma, x_\theta + \sigma)$. Then, the optimal x trades off between setting the derivative of the ψ terms equal to 0and the $(1-\psi)$ terms equal to 0. Crucially, if $\frac{\beta_1 - 2\delta x_\theta}{2\mathbb{E}[\gamma|\delta]} = \sigma$, and $(x_\theta - x^*) = \sigma^2$, there is no tradeoff. As σ grows arbitrarily large, x^* moves further away from $x^\dagger$ and closer to $x_\theta \pm \sigma$.

Next, applying the envelope theorem to (8) with respect to σ^2 gives

$$(1-\psi)(\mathbb{E}[\gamma|\delta] - \delta) + 2\psi[(x^* - x_\theta)^2 - \sigma^2]\mathrm{Var}(\gamma|\delta) \quad (9)$$

Denoting the maximand of (8) by x^*, using the envelope theorem, the derivative of V^F with respect to σ^2 is

$$(1-\psi)(\mathbb{E}[\gamma|\delta] - \delta) + 2\psi[(x^* - x_\theta)^2 - \sigma^2]\mathrm{Var}(\gamma|\delta) \quad (10)$$

Returning to (8) and differentiating with respect to x, x^* is defined by

$$\begin{aligned} (1-\psi)\big(\beta_1 - 2\delta x^* - 2(\mathbb{E}[\gamma|\delta] - \delta)(x^* - x_\theta)\big) + 4\psi\mathrm{Var}(\gamma|\delta)(x^* - x_\theta)\big(\sigma^2 - (x^* - x_\theta)^2\big) = 0 \\ \implies \sigma^2 - (x^* - x_\theta)^2 = \frac{(1-\psi)\big(2(\mathbb{E}[\gamma|\delta] - \delta)(x^* - x_\theta) + 2\delta x^* - \beta_1\big)}{4\psi\mathrm{Var}(\gamma|\delta)(x^* - x_\theta)} \end{aligned}$$

in the case where $x^* \neq x_\theta$. This means (10) is ≥ 0 if and only if

$$(1-\psi)(\mathbb{E}[\gamma|\delta]-\delta) \geq 2\psi[\frac{(1-\psi)\big(2(\mathbb{E}[\gamma|\delta]-\delta)(x^*-x_\theta)+2\delta x^*-\beta_1\big)}{4\psi \mathrm{Var}(\gamma|\delta)(x^*-x_\theta)}]\mathrm{Var}(\gamma|\delta)$$
$$(1-\psi)(\mathbb{E}[\gamma|\delta]-\delta) \geq (1-\psi)(\mathbb{E}[\gamma|\delta]-\delta)+\psi\frac{2\delta x^*-\beta_1}{x^*-x_\theta}$$
$$\frac{\beta_1-2\delta x^*}{x^*-x_\theta} \geq 0.$$

This expression allows us to derive the final two results. First, if $x^\dagger \geq x_\theta$, then x^* is between $x^\dagger$ and $x_\theta+\sigma$, and is further attracted to the latter point as σ grows arbitrarily large. In this case, since $\delta > 0$, $f'_S(x^*) = \beta_1 - 2\delta x^* < 0$. A similar argument applies when $x^\dagger < x_\theta$. Thus, for σ arbitrarily large, V^F is decreasing in σ. This gives the second result.

For the third result, consider $\sigma \approx 0$. Suppose $\beta_1 - 2\delta x_\theta = 0$; this means that $x^\dagger = x_\theta$. Hence, for any $\sigma \geq 0$, x^* lies weakly between x_θ and $x_\theta+\sigma$ (strictly if $\sigma > 0$). Since $\beta_1 - 2\delta x_\theta = 0$, we have $\beta_1 - 2\delta x^* < 0$, which means V^F is decreasing in σ.[1]

Finally, consider again $\sigma \approx 0$ and suppose $\beta_1 - 2\delta x_\theta > 0$. Note that this implies $x^\dagger > x_\theta$. However, as $\psi \to 1$, $x^* \to x_\theta$. Since $\beta_1 - 2\delta x_\theta > 0$, this implies that for ψ large, $\beta_1 - 2\delta x^* > 0$, as well, meaning the derivative is positive. An analogous argument applies when $\beta_1 - 2\delta x_\theta < 0$. In either case, V^F is increasing in σ for σ small and $\beta_1 - 2\delta x_\theta \neq 0$. □

Proposition 4. *Consider two types of senders $\theta = [\underline{\delta}, \overline{\delta}]$ and $[\underline{\delta}', \overline{\delta}']$ such that $\theta \cap \theta' \neq \varnothing$ and $\theta \cap \theta' \subseteq \theta^\circ$. Suppose R receives information from either sender with positive probability. Fix a sender model $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$ such that $\delta \in \theta \cap \theta'$. Given $f_S(x)$, a type θ sender's utility strictly increases (decreases) from the existence of type θ' if one of the following hold.*

1. *$\delta \in \theta^\circ \neq \varnothing$, $\delta = \overline{\delta}'$, and θ' distantly (locally) samples.*
2. *$\delta \in \theta^\circ \neq \varnothing$, $\delta = \underline{\delta}'$, and θ' locally (distantly) samples.*
3. *$\delta = \overline{\delta}$, $\theta^\circ = \varnothing$, type θ locally samples, and θ' distantly (locally) samples.*
4. *$\delta = \underline{\delta}$, $\theta^\circ = \varnothing$, type θ distantly samples, and θ' locally (distantly) samples.*
5. *$\theta^\circ = \varnothing$, $\theta'^\circ \neq \varnothing$, $\delta = \overline{\delta}'$, and θ' distantly (locally) samples.*
6. *$\theta^\circ = \varnothing$, $\theta'^\circ \neq \varnothing$, $\delta = \underline{\delta}'$, and θ' locally (distantly) samples.*

Proof. First, we show how the receiver's expected value from entertaining models that have steeper or shallower tradeoffs than the sender's affect her maximal expected utility. Suppose $\gamma > \delta$. Given

[1] x^* also has a symmetric solution that lies between x_θ and $x_\theta - \sigma$, in which case the logic is exactly the same.

a sampling strategy σ, we have:

$$\max \beta_0 + \beta_1 x - \delta(x - x_\theta)^2 = \beta_0 + \beta_1 x_\theta + \frac{\beta_1^2}{4\delta}$$
$$\max \beta_0 + \beta_1 x - \gamma(x - x_\theta)^2 + (\gamma - \delta)\sigma^2 = \beta_0 + \beta_1 x_\theta + \frac{\beta_1^2}{4\gamma} + (\gamma - \delta)\sigma^2.$$

The latter is greater than the former when

$$\frac{\beta_1^2}{4\gamma} + (\gamma - \delta)\sigma^2 \geq \frac{\beta_1^2}{4\delta} \iff (\gamma - \delta)\sigma^2 \geq \frac{\beta_1^2}{4}\left(\frac{1}{\delta} - \frac{1}{\gamma}\right) \iff \sigma^2 \geq \frac{\beta_1^2}{\gamma\delta}.$$

Now, suppose $\delta \in \theta_i^\circ, \theta_j^\circ$. Then, the receiver entertains just the sender's model, which she believes is the sender's (only) model. Thus, we consider situations where the boundaries of the sets intersect.

Next, suppose $\underline{\delta}_i \in \theta_j^\circ$. Conditional on seeing $\underline{\delta}_i$, the receiver entertains models with $\gamma \leq \underline{\delta}_i$. By Theorem 1, the sender would receive to locally sample in these cases. However, in her initial choice of σ, the sender is forced to consider both the probability of $\overline{\delta}_i$ and $\underline{\delta}_i$. If $\sigma_i = \overline{\sigma}$, the sender is being forced to distantly sample, which delivers a gain most of the time but a loss in the case where $\underline{\delta}_i$ materializes. However, if $\underline{\delta}_i \in \theta_j^\circ$, then with some probability, the receiver believes the $\underline{\delta}_i$ model was the true model and came from the type j sender. This means the receiver places a higher weight on the sender's model itself, and less on models with lower γ values (which, under $\sigma_i = \overline{\sigma}$, deliver lower expected utility).

A similar argument shows that the sender *loses out* when $\delta = \underline{\delta} - i$ and $\sigma_i = \underline{\sigma}$. In this case, the sender is (optimally) gaining by herself locally sampling, meaning the remaining models the receiver entertains deliver higher expected utility. However, the receiver thinks with positive probability that the sender's model is in fact the *true* model, causing her to relatively downweight these alternative models and focus more weight on the sender's model, which delivers lower expected utility. Yet another similar arugment shows that the sender gains (loses) when $\overline{\delta}_i \in \theta_j^\circ$ and $\sigma_i = \underline{\sigma}$ ($\sigma_i = \overline{\sigma}$).

Next, suppose $\delta \in \theta_i^\circ$ and that $\delta = \overline{\delta}_j$. If the type j sender is distantly sampling ($\sigma_j = \overline{\sigma}$), the receiver entertains not only type i's model but also models with higher γ (since she believes the model could have also come from θ_j). The set of models R entertains is thus rosier than if she were to just entertain the type i sender's model, meaning the type i sender benefits from the type j's distant sampling Local sampling, on the other hand, would hurt her. A similar argument takes cares of the case where $\delta = \underline{\delta}_j$, where the type θ_i sender is hurt by distant sampling and benefits from local sampling.

□

Theorem 2. *Suppose P_γ^D is continuous. Then, the optimal type space is the collection of singletons on $[0, \overline{\gamma}]$: $\Theta = \{\{\delta\} : \delta \in \Gamma\}$. Moreover, T_θ places positive mass on all points with positive mass in P_γ^R. D's utility is increasing in q_0.*

Proof. Formal proof under construction, see broad sketches of pieces below.

First we show that the optimal type space contains no non-degenerate intervals. Suppose by contradiction that there existed an interval of type $\theta = [\underline{\delta}_\theta, \overline{\delta}_\theta]$ with $\theta \subseteq \Gamma^\circ$. Suppose $\theta^\circ \neq \varnothing$ and consider $\delta \in \theta^\circ$. Conditional on seeing this model, the sender believes a model of the form $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$ is the unique model.

Now, for each each $\delta \in \theta^\circ$, D can create a type $\{\delta\}$. This type strategically samples by setting σ such that $\max_x f_S(x) \leq \max_x f_S(x) + (\mathbb{E}[\gamma|\delta])(\sigma^2 - (x - x_\theta)^2)$. Thus, conditional on seeing a model of the form $\beta_0 + \beta_1 x - \delta x^2$, R believes that $f_S(x)$ is either the true model or the model from the $\{\delta\}$ type. Let $q(\delta)$ be the conditional probability of the type $\{\delta\}$ sender relative to the original type θ sender. Then, this generates a difference on the order of

$$\int_{\underline{\delta}_\theta}^{\overline{\delta}_\theta} \max_x f_S(x) + q(\delta)\big(\overline{\gamma}_{\{\delta\}} - \delta\big)(\overline{\sigma}^2 - (x - x_\theta)^2) - \max_x f_S(x) dP_\gamma^D(\delta \leq \mathbb{E}_\gamma^R[\gamma] | \delta \in \theta^\circ)$$
$$+ \int_{\underline{\delta}_\theta}^{\overline{\delta}_\theta} \max_x f_S(x) + q(\delta)\big(\underline{\gamma}_{\{\delta\}} - \delta\big)(\underline{\sigma}^2 - (x - x_\theta)^2) - \max_x f_S(x) dP_\gamma^D(\delta > \mathbb{E}_\gamma^R[\gamma] | \delta \in \theta^\circ)$$

The first line considers any arbitrary senders who distantly sample. The second line considers any arbitrary senders who locally sample. In both cases, the gain is positive and is increasing in $q(\delta)$; i.e., the gain is higher the more likely R is to think that $f_S(x)$ came from type $\{\delta\}$.

We are left to consider the boundary types of θ: $\overline{\delta}^\theta$ and $\underline{\delta}^\theta$. Since θ eitehr locally or distantly samples, by Proposition 4 one of $\overline{\delta}^\theta$ or $\underline{\delta}^\theta$ gains and one loses from the existence of an arbitrary type. However, because these singletons have measure 0 in the prior, replacing each with an arbitrary type has no effect on expected utility. Thus, the optimal type space contains no intervals, and is only composed of singletons. By the first part of the result, this set of singletons must cover all of Γ, meaning $\Theta = \{\{\delta\} : \delta \in \Gamma\}$ is the optimal set of types. Returning to D's ex-ante utility, S chooses the density of each $\{\delta\}$ to be precisely the density in the prior. □

Proposition 5. *Suppose the type space designer's prior P_γ^D over Γ is discrete. Then the optimal type space is given by a partition of Γ into a combination of singletons $\{\delta_i\}$ and doubles $\{\delta_i, \delta_{i+1}\}$.*

Proof. Formal proof under construction. Sketch is as follows. The logic for the proof with discrete types is largely the same, except that we have type pooling. Consider two γ_i values such that γ_i, γ_{i+1} are next to each other. We either under or overstate tradeoffs (suppose wlog that we overstate). This generates a "loss" from γ_i and a "gain" for γ_{i+1}. This ensures we only entertain models with $\gamma \leq \gamma_{i+1}$ instead of $\gamma \neq \gamma_{i+1}$ (which are still on average "above" γ, just by less. If the weight on γ_i grows vanishingly small, it's optimal to pool these two together.

To see that pooling involves at most two types, note that if there were three types, the "middle" type would always be the true model if it realized. So we are better off either pooling two together and "breaking out" the third or breaking out all three.

□